

Clustering Countries Based on Happiness Indicators

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Sections

- Introduction
- Data Inspection & Preparation
- Exploratory Data Analysis
- Models
- Additional Work
- Conclusion

Introduction

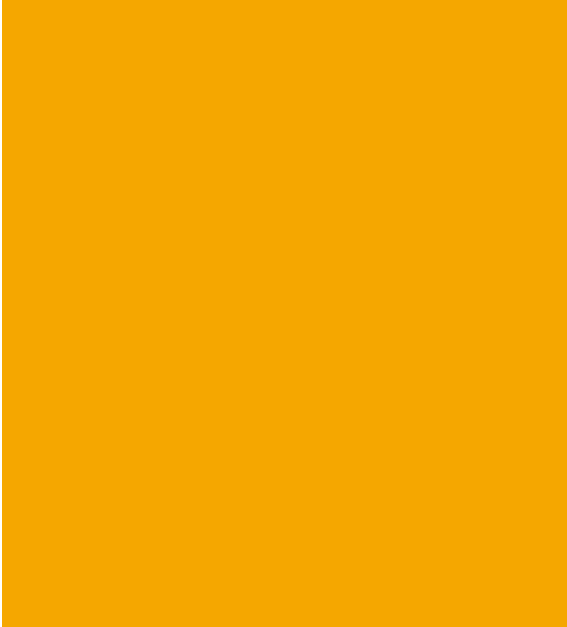
This project explores how countries differ in terms of overall happiness and the factors that contribute to it. We want to find meaningful patterns in the data to see whether countries naturally group together based on indicators like GDP per capita, social support, life expectancy, freedom to make life choices, generosity, and perceptions of corruption.

What We Want to Achieve:

- **Segment Countries:** We are using unsupervised learning techniques, which means we don't tell the algorithm what groups exist ahead of time. Instead, it discovers groups or "clusters" based on similarities in the data.
- **Policy and Research Insights:** By determining these clusters, we might identify groups of countries with similar strengths and weaknesses, which could inform policy decisions and further research on global well-being.
- **Methodological Exploration:** The project also considers using advanced methods such as robust distance measures, multiple clustering techniques, and dimensionality reduction to help interpret the hidden structure of the data.



Data Inspection



Datasets

The dataset comes from the World Happiness Report. Across different years (2015 to 2019 in your case), the reports include:

Happiness Scores: An overall rating of happiness for each country or region.

Contributing Factors: Values associated with various determinants of happiness, such as:

- **Economy (or GDP per capita):** A measure of the standard of living.
- **Social Support:** Reflects the support people receive from family, friends, and community.
- **Life Expectancy/Healthy Life Expectancy:** An indicator of health.
- **Freedom to Make Life Choices:** How much personal freedom exists in each country.
- **Generosity:** Levels of charitable giving or altruism.
- **Perceptions of Corruption:** How corrupt people feel their government or institutions are.
- **Other Metrics:** Some years include additional measures such as the "Dystopia Residual" or confidence intervals.

Dataset Overview (2015)

Country	Region	Happiness Rank	Happiness Score	Standard Error	Economy (GDP per Capita)	Family	Health (Life Expectancy)	Freedom	Trust (Government Corruption)	Generosity	Dystopia Residual
Syria	Middle East and Northern Africa	156	3.006	0.05015	0.66320	0.47489	0.72193	0.15684	0.18906	0.47179	0.32858
Hong Kong	Eastern Asia	72	5.474	0.05051	1.38604	1.05818	1.01328	0.59608	0.37124	0.39478	0.65429
Rwanda	Sub-Saharan Africa	154	3.465	0.03464	0.22208	0.77370	0.42864	0.59201	0.55191	0.22628	0.67042
Sri Lanka	Southern Asia	132	4.271	0.03751	0.83524	1.01905	0.70806	0.53726	0.09179	0.40828	0.67108
Bulgaria	Central and Eastern Europe	134	4.218	0.04828	1.01216	1.10614	0.76649	0.30587	0.00872	0.11921	0.89991
Cambodia	Southeastern Asia	145	3.819	0.05069	0.46038	0.62736	0.61114	0.66246	0.07247	0.40359	0.98195
Gabon	Sub-Saharan Africa	143	3.896	0.04547	1.06024	0.90528	0.43372	0.31914	0.11091	0.06822	0.99895
Dominican Republic	Latin America and Caribbean	98	4.885	0.07446	0.89537	1.17202	0.66825	0.57672	0.14234	0.21684	1.21305
Hungary	Central and Eastern Europe	104	4.800	0.06107	1.12094	1.20215	0.75905	0.32112	0.02758	0.12800	1.24074
Portugal	Western Europe	88	5.102	0.04802	1.15991	1.13935	0.87519	0.51469	0.01078	0.13719	1.26462

158 rows and 12 columns

Dataset Statistics (2015)

- **Happiness Rank and Score:** The ranking confirms a full ordered list from the best (rank 1) to the worst (rank 158). The happiness score has a median of about 5.232 and a mean of 5.376, indicating that many countries fall in the middle range with fewer extremes at either end.
- **Economy (GDP per Capita)** shows moderate variability (median ≈ 0.910), suggesting economic performance differences among countries.
- **Family and Social Support** components are relatively high for most countries, given that the median family value is about 1.0295.
- **Health (Life Expectancy)** has a median around 0.6967, with some countries reaching above 1, indicating better overall health conditions.

Dataset Statistics (2015)

- **Freedom** has a median around 0.4355, which points to moderately varying levels of personal freedom.
- **Trust (Government Corruption)** has a median of roughly 0.107, suggesting that many countries report lower trust levels.
- **Generosity** varies moderately with a median of 0.216, while its spread indicates some countries have much higher levels of perceived generosity.
- **Dystopia Residual**: This value is used as a benchmark or “baseline” in the Happiness Report. In the 2015 data, the residuals range from approximately 0.329 to 3.602, with a median around 2.095. This spread highlights the portion of happiness that isn’t explained by the standard factors measured.

Dataset Overview (2016)

Country	Region	Happiness Rank	Happiness Score	Lower Confidence Interval	Upper Confidence Interval	Economy (GDP per Capita)	Family	Health (Life Expectancy)	Freedom	Trust (Government Corruption)	Generosity	Dystopia Residual
Denmark	Western Europe	1	7.526	7.460	7.592	1.44178	1.16374	0.79504	0.57941	0.44453	0.36171	2.73939
Switzerland	Western Europe	2	7.509	7.428	7.590	1.52733	1.14524	0.86303	0.58557	0.41203	0.28083	2.69463
Iceland	Western Europe	3	7.501	7.333	7.669	1.42666	1.18326	0.86733	0.56624	0.14975	0.47678	2.83137
Norway	Western Europe	4	7.498	7.421	7.575	1.57744	1.12690	0.79579	0.59609	0.35776	0.37895	2.66465
Finland	Western Europe	5	7.413	7.351	7.475	1.40598	1.13464	0.81091	0.57104	0.41004	0.25492	2.82596
Canada	North America	6	7.404	7.335	7.473	1.44015	1.09610	0.82760	0.57370	0.31329	0.44634	2.70485
Netherlands	Western Europe	7	7.339	7.284	7.394	1.46468	1.02912	0.81231	0.55211	0.29927	0.47416	2.70749
New Zealand	Australia and New Zealand	8	7.334	7.264	7.404	1.36066	1.17278	0.83096	0.58147	0.41904	0.49401	2.47553
Australia	Australia and New Zealand	9	7.313	7.241	7.385	1.44443	1.10476	0.85120	0.56837	0.32331	0.47407	2.54650
Sweden	Western Europe	10	7.291	7.227	7.355	1.45181	1.08764	0.83121	0.58218	0.40867	0.38254	2.54734

157 rows and 13 columns

Dataset Statistics (2016)

- **Happiness Score:**
 - **Range:** The scores vary from a minimum of 2.905 to a maximum of 7.526.
 - **Central Tendency:** The median score is around 5.314, with a mean of approximately 5.382. This suggests that most countries fall near the mid-range of the score distribution, with only a few achieving very high happiness scores.
- **Confidence Intervals:** The dataset now includes both lower (min $\approx$ 2.732, 1st quartile $\approx$ 4.327, median $\approx$ 5.237, max $\approx$ 7.460) and upper confidence intervals (min $\approx$ 3.078, 1st quartile $\approx$ 4.465, median $\approx$ 5.419, max $\approx$ 7.669). These provide an estimate of the uncertainty around the happiness scores and can be useful later to assess the reliability of the reported scores.
- **Economy (GDP per Capita):** Values range from 0 up to approximately 1.8243, with a median near 1.0278. This variability indicates different levels of economic conditions among countries.

Dataset Statistics (2016)

- **Family:** Values range from 0 to around 1.1833, with a median of about 0.8414.
- **Health (Life Expectancy):** Ranges from 0 up to around 0.9528, with a median near 0.5966.
- **Freedom:** Varies between 0 and about 0.6085, with a median of approximately 0.3975.
- **Trust (Government Corruption):** Ranges from 0 up to around 0.50521, with a median of roughly 0.1055.
- **Generosity:** Ranges from 0 to a maximum of about 0.8197, with a median near 0.2225.
- **Dystopia Residual:** This metric ranges from around 0.8179 up to 3.8377, with a median of about 2.2907. It represents the portion of the happiness score not explained by the main factors, serving as a baseline comparison.

Dataset Overview (2017)

Country	Happiness.Rank	Happiness.Score	Whisker.high	Whisker.low	Economy..GDP.per.Capita.	Family	Health..Life.Expectancy.	Freedom	Generosity	Trust..Government.Corruption.	Dystopia.Residual
Norway	1	7.537	7.594445	7.479556	1.61646318	1.5335236	0.79666650	0.63542259	0.36201224	0.315963835	2.2770267
Denmark	2	7.522	7.581728	7.462272	1.48238301	1.5511216	0.79256552	0.62600672	0.35528049	0.400770068	2.3137074
Iceland	3	7.504	7.622030	7.385970	1.48063302	1.6105740	0.83355212	0.62716264	0.47554022	0.153526559	2.3227153
Switzerland	4	7.494	7.561772	7.426227	1.56497955	1.5169117	0.85813129	0.62007058	0.29054928	0.367007285	2.2767162
Finland	5	7.469	7.527542	7.410458	1.44357193	1.5402467	0.80915767	0.61795086	0.24548277	0.382611543	2.4301815
Netherlands	6	7.377	7.427426	7.326574	1.50394464	1.4289392	0.81069613	0.58538449	0.47048983	0.282661825	2.2948041
Canada	7	7.316	7.384403	7.247597	1.47920442	1.4813490	0.83455765	0.61110091	0.43553972	0.287371516	2.1872644
New Zealand	8	7.314	7.379510	7.248490	1.40570605	1.5481951	0.81675971	0.61406213	0.50000513	0.382816702	2.0464563
Sweden	9	7.284	7.344095	7.223905	1.49438727	1.4781622	0.83087516	0.61292410	0.38539925	0.384398729	2.0975380
Australia	10	7.284	7.356651	7.211349	1.48441494	1.5100420	0.84388679	0.60160738	0.47769925	0.301183730	2.0652108

155 rows and 12 columns

Dataset Statistics (2017)

Happiness Score:

- **Range:** Approximately 2.693 to 7.537.
- **Central Tendency:** The median is about 5.279 and the mean is 5.354, which suggests that while there are some very high scores, the distribution of happiness scores is centered around the mid-range.

Confidence Intervals (Whiskers):

- **Whisker.high:** Median near 5.370, with maximum values approaching 7.622.
- **Whisker.low:** Median around 5.193, which combined with the high whiskers provides insight into the variability and reliability of the reported happiness scores.

Economy (GDP per Capita):

- The median is approximately 1.0646 with a mean of 0.9847, indicating that most countries fall in a moderately narrow economic performance range, although the maximum reaches about 1.87.

Dataset Statistics (2017)

- **Family:** Has a median value of around 1.254, suggesting relatively high family support on average.
- **Health (Life Expectancy):** Median is about 0.6060, indicating the general level of health and lifespan.
- **Freedom:** With a median around 0.4375, there is moderate variation in perceived freedom among countries.
- **Generosity:** The median is about 0.2315, and values vary, showing differences in how generosity is exhibited across societies.
- **Trust (Government Corruption):** The median is roughly 0.08985 with a maximum near 0.464, signifying that while many countries report low trust levels, there is still notable variation.
- **Dystopia Residual:** This metric has a median of about 1.8329 and a mean of 1.8502, with values ranging from around 0.378 to 3.118. It represents the baseline level of unhappiness not accounted for by the measured factors and is crucial for understanding the unexplained variance in overall happiness.

Dataset Overview (2018)

Overall rank	Country or region	Score	GDP per capita	Social support	Healthy life expectancy	Freedom to make life choices	Generosity	Perceptions of corruption
1	Finland	7.632	1.305	1.592	0.874	0.681	0.202	0.393
2	Norway	7.594	1.456	1.582	0.861	0.686	0.286	0.340
3	Denmark	7.555	1.351	1.590	0.868	0.683	0.284	0.408
4	Iceland	7.495	1.343	1.644	0.914	0.677	0.353	0.138
5	Switzerland	7.487	1.420	1.549	0.927	0.660	0.256	0.357
6	Netherlands	7.441	1.361	1.488	0.878	0.638	0.333	0.295
7	Canada	7.328	1.330	1.532	0.896	0.653	0.321	0.291
8	New Zealand	7.324	1.268	1.601	0.876	0.669	0.365	0.389
9	Sweden	7.314	1.355	1.501	0.913	0.659	0.285	0.383
10	Australia	7.272	1.340	1.573	0.910	0.647	0.361	0.302

Dataset Statistics (2018)

Overall rank: Ranges from 1 to 156, with a median and mean of 78.50. This indicates the dataset includes the full spectrum of country rankings.

Happiness Score ("Score"): Minimum value is 2.905 and the maximum is 7.632. The median score is 5.378, and the mean is about 5.376, showing that most countries fall in the mid-range of happiness.

GDP per capita: Values range from 0.0000 to 2.0960, with a 1st quartile of 0.6162, a median of 0.9495, and a mean of 0.8914. This wide range indicates considerable differences in economic performance across countries.

Social support: Ranges from 0.000 to 1.644, with a median of 1.255 and a mean of 1.213, suggesting that while many countries report high levels of support, there is still notable variability.

Dataset Statistics (2018)

Healthy life expectancy:

- Values span from 0.0000 to 1.0300.
- The median is 0.6440 and the mean is 0.5973, indicating moderate differences in expected healthy years across countries.

Freedom to make life choices:

- With values ranging from 0.0000 to 0.7240, the median is 0.4870 and the mean is 0.4545.
- This reflects variability in perceived personal freedom across nations.

Generosity:

- Ranges from 0.0000 to 0.5980, with a median of 0.1740 and a mean of 0.1810, showing generally low-to-moderate levels of generosity across countries.

Perceptions of corruption:

- This variable is stored as text, so a numerical summary isn't directly provided. It represents the level of perceived corruption and would need conversion to numeric for quantitative analysis.

Dataset Overview (2019)

Overall rank	Country or region	Score	GDP per capita	Social support	Healthy life expectancy	Freedom to make life choices	Generosity	Perceptions of corruption
1	Finland	7.769	1.340	1.587	0.986	0.596	0.153	0.393
2	Denmark	7.600	1.383	1.573	0.996	0.592	0.252	0.410
3	Norway	7.554	1.488	1.582	1.028	0.603	0.271	0.341
4	Iceland	7.494	1.380	1.624	1.026	0.591	0.354	0.118
5	Netherlands	7.488	1.396	1.522	0.999	0.557	0.322	0.298
6	Switzerland	7.480	1.452	1.526	1.052	0.572	0.263	0.343
7	Sweden	7.343	1.387	1.487	1.009	0.574	0.267	0.373
8	New Zealand	7.307	1.303	1.557	1.026	0.585	0.330	0.380
9	Canada	7.278	1.365	1.505	1.039	0.584	0.285	0.308
10	Austria	7.246	1.376	1.475	1.016	0.532	0.244	0.226

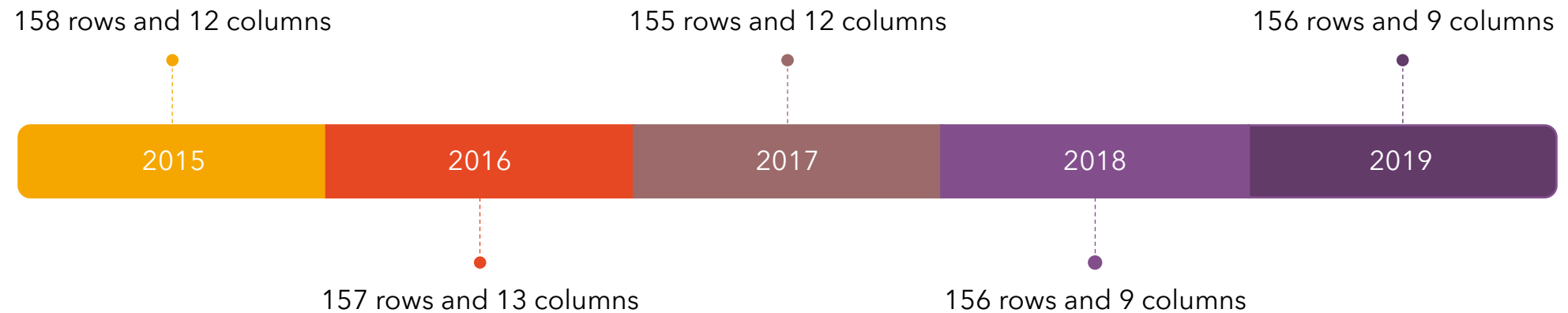
Dataset Statistics (2019)

- **Overall Rank:** Ranges from 1 to 156, with a mean of 78.50. This indicates a complete ranking from the highest to the lowest positions.
- **Score:** Ranges from a minimum of 2.853 to a maximum of 7.769, with a median near 5.380 and a mean of approximately 5.407. This spread indicates that while a few countries score very high, many countries are clustered towards the middle of the happiness spectrum.
- **GDP per Capita:** Varies between 0 (minimum) and about 1.684, with the median around 0.960. A few higher values (near 1.684) suggest significant differences in economic performance.
- **Social Support:** The values range roughly between 0 and 1.624 (with a median of 1.272 and mean around 1.209), implying variation in the support systems available across countries.

Dataset Statistics (2019)

- **Healthy Life Expectancy:** Has a tight range from the minimum up to about 1.141 with a median of 0.789 and mean of around 0.725, suggesting most values are within a moderately narrow interval.
- **Freedom to Make Life Choices:** With values ranging from 0 up to about 0.631 (median of 0.417, mean 0.3926), there appears to be moderate variability in perceived freedom.
- **Generosity:** The values range from 0 to 0.566 (median 0.1775, mean 0.1848), indicating a moderate spread in how generosity is perceived.
- **Perceptions of Corruption:** Range from 0 to 0.453 (median 0.0855, mean 0.1106). The lower medians relative to the range suggest that many countries report low levels of perceived corruption, though some report much higher perceptions.

Dimension of Datasets by Year





Data Preparation



Merging and Harmonization of Datasets

Process

- Merging the datasets from 2015, 2016, 2017, 2018, and 2019
- We standardizing variable names by renaming columns
- Adding a “Year” column to each dataset to track the time dimension
- Selecting only the common variables across the datasets so that all data could be compared directly
- Finally, Creating a consistent panel dataset covering multiple years
- There was only one missing value which I have imputed using median

Benefits

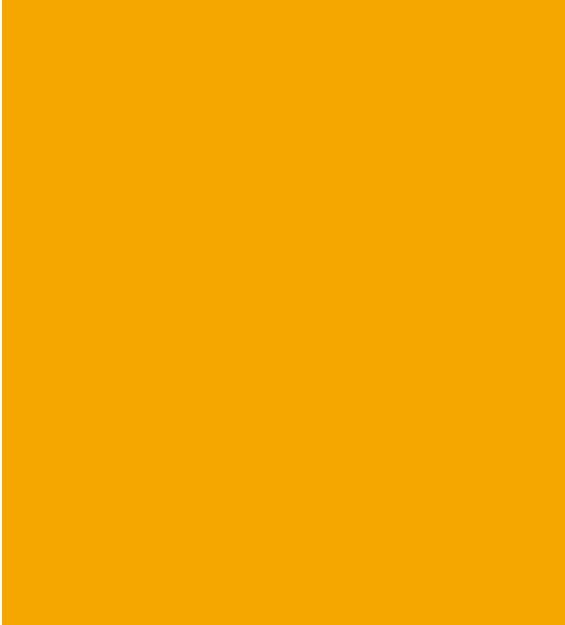
- The unified panel data allows comparing country performance across time
- Increasing sample size, which improved the reliability of our analysis
- Harmonizing the data makes it easier to apply the same analytical methods (such as clustering or dimensionality reduction) across all years
- A consistent structure helps explore trends and changes in happiness determinants over time

Final Dataset (Panel Dataset)

Country	Year	Score	Social_support	Healthy_life_expectancy	Freedom	Generosity	Corruption	GDP_per_capita
Ivory Coast	2015	3.655	0.77115	0.15185	0.46866	0.20165	0.17922	0.46534
Burkina Faso	2015	3.587	0.85188	0.27125	0.39493	0.21747	0.12832	0.25812
Afghanistan	2015	3.575	0.30285	0.30335	0.23414	0.36510	0.09719	0.31982
Rwanda	2015	3.465	0.77370	0.42864	0.59201	0.22628	0.55191	0.22208
Benin	2015	3.340	0.35386	0.31910	0.48450	0.18260	0.08010	0.28665
Syria	2015	3.006	0.47489	0.72193	0.15684	0.47179	0.18906	0.66320
Burundi	2015	2.905	0.41587	0.22396	0.11850	0.19727	0.10062	0.01530
Togo	2015	2.839	0.13995	0.28443	0.36453	0.16681	0.10731	0.20868
Denmark	2016	7.526	1.16374	0.79504	0.57941	0.36171	0.44453	1.44178
Switzerland	2016	7.509	1.14524	0.86303	0.58557	0.28083	0.41203	1.52733
Iceland	2016	7.501	1.18326	0.86733	0.56624	0.47678	0.14975	1.42666
Norway	2016	7.498	1.12690	0.79579	0.59609	0.37895	0.35776	1.57744
Finland	2016	7.413	1.13464	0.81091	0.57104	0.25492	0.41004	1.40598
Canada	2016	7.404	1.09610	0.82760	0.57370	0.44834	0.31329	1.44015
Netherlands	2016	7.339	1.02912	0.81231	0.55211	0.47416	0.29927	1.46468

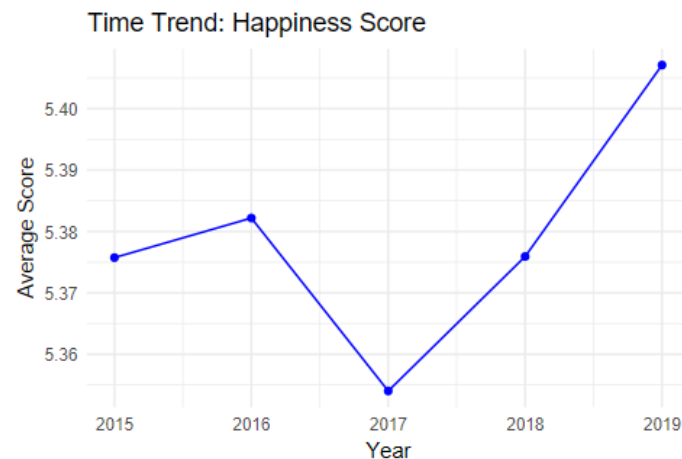


Exploratory Data Analysis



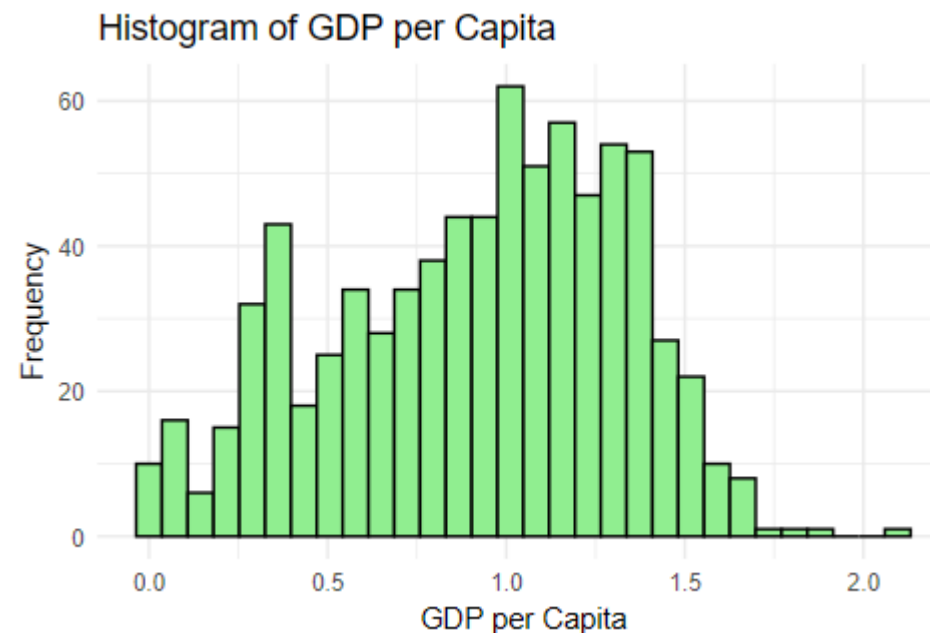
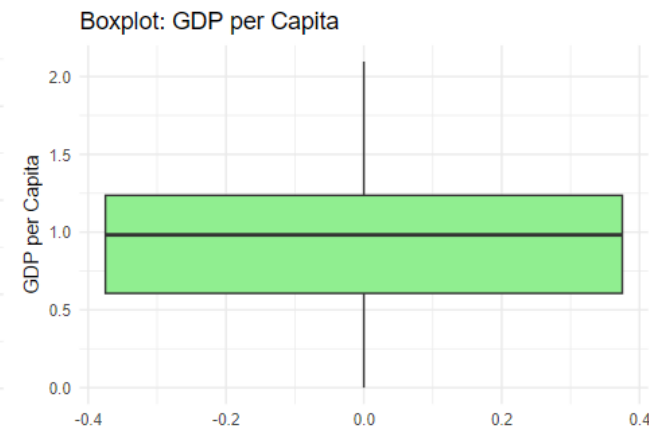
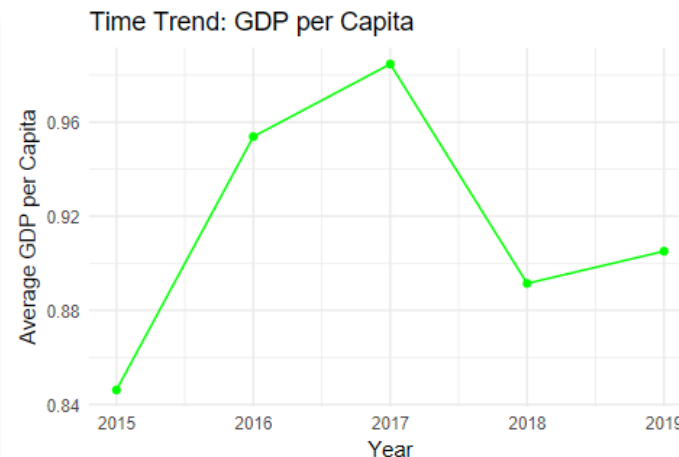
Happiness Score

- unusual dip around 2017
- increase in 2019
- no extreme outliers
- skewed but not severely



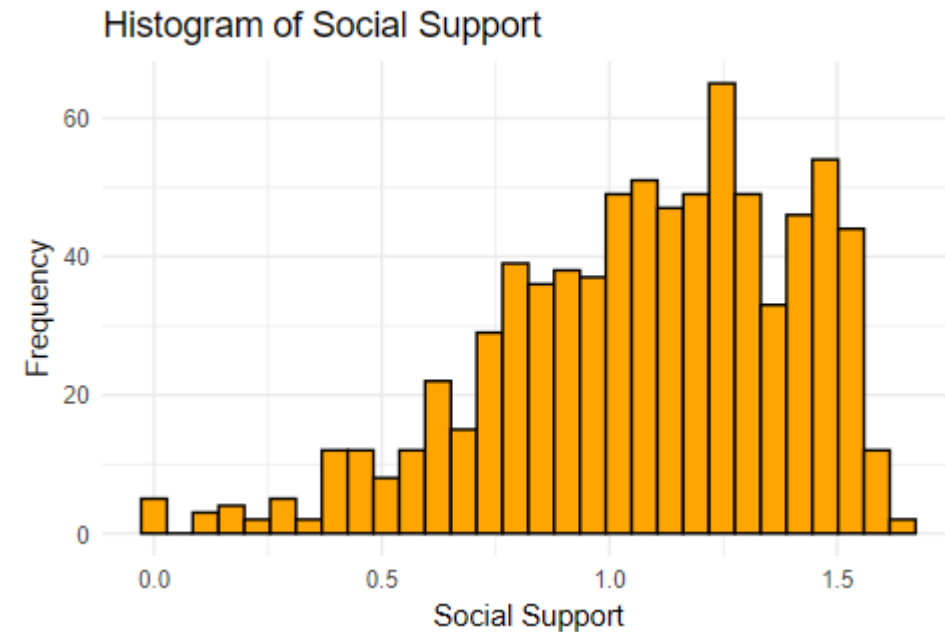
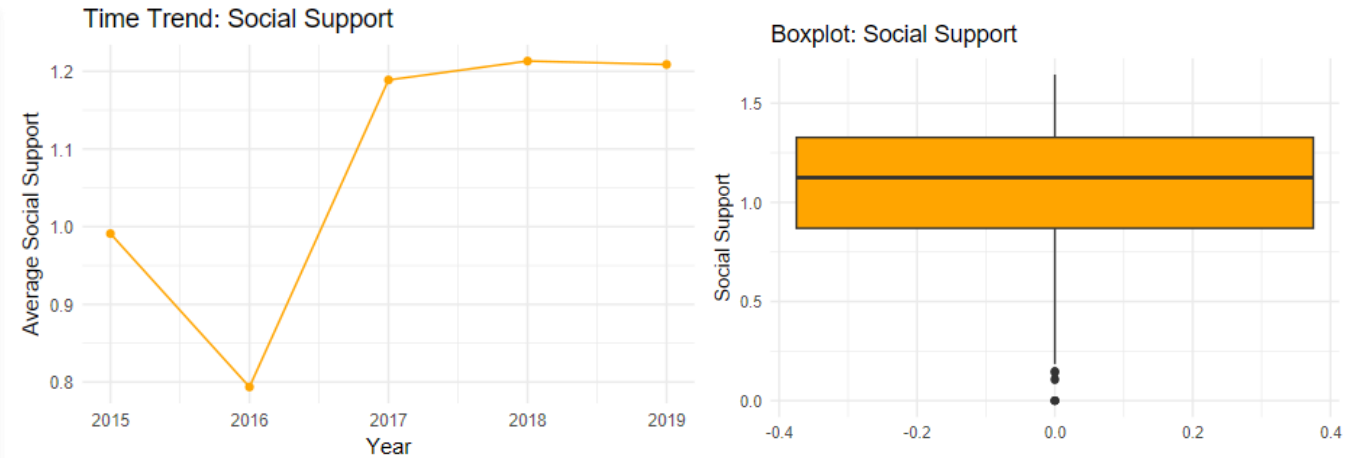
GDP per Capita

- Significant increase from 2015 to 2017
- A dip on 2018
- higher-end outliers are visible
- Transformation: A logarithmic transform often helps when GDP is highly skewed, so it might reduce the impact of high-GDP outliers during clustering.
- Outlier Handling: In robust clustering, you might trim or downweight extremely wealthy countries if they distort clusters.



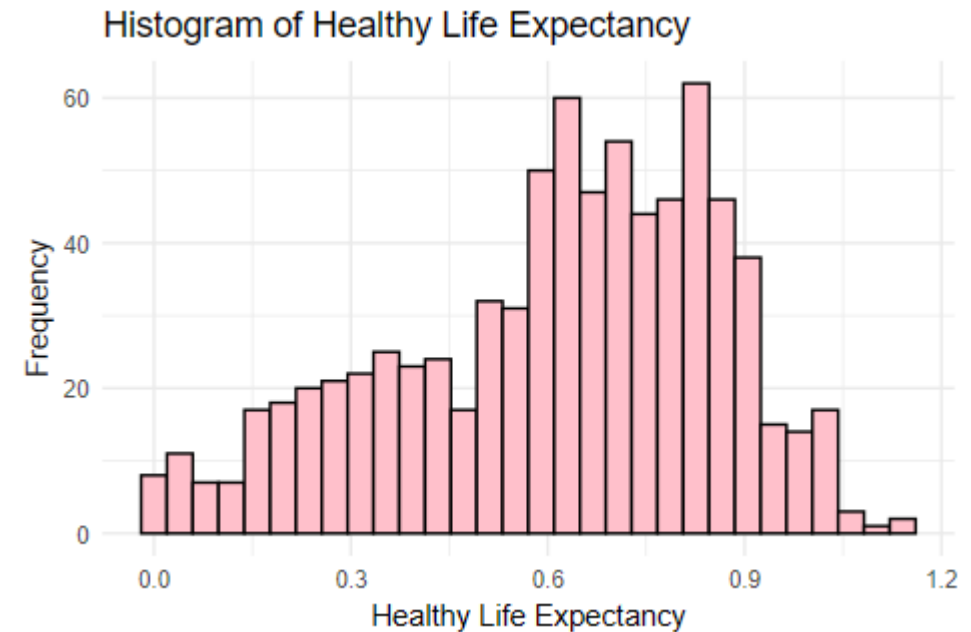
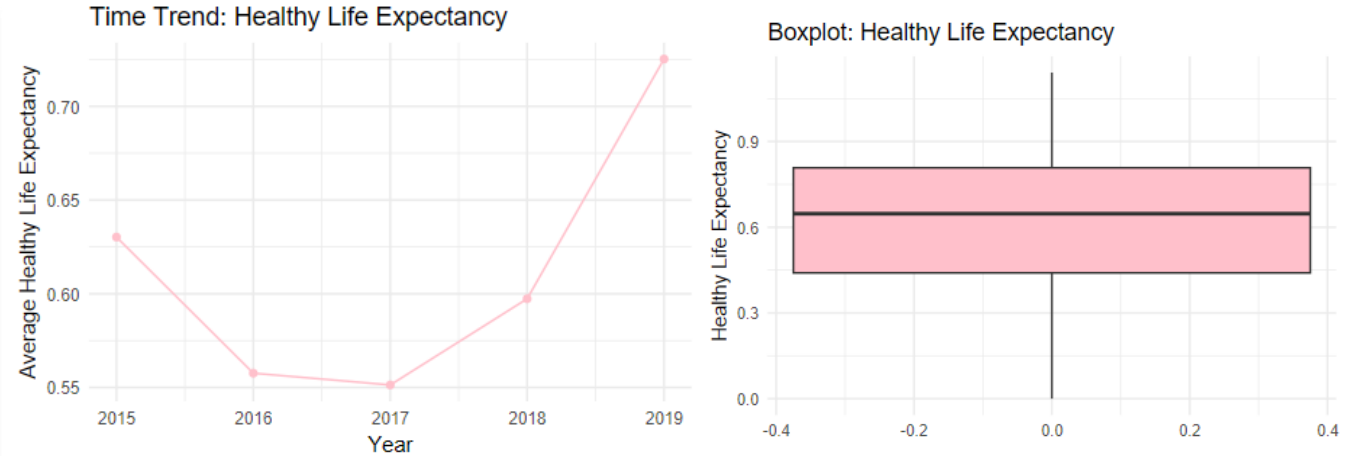
Social Support

- drop in 2016 followed by a strong rebound
- mild outlier
- Because Social Support is typically high for most countries, differences might be subtle. Standardizing or normalizing could help the algorithm better discriminate among smaller differences.



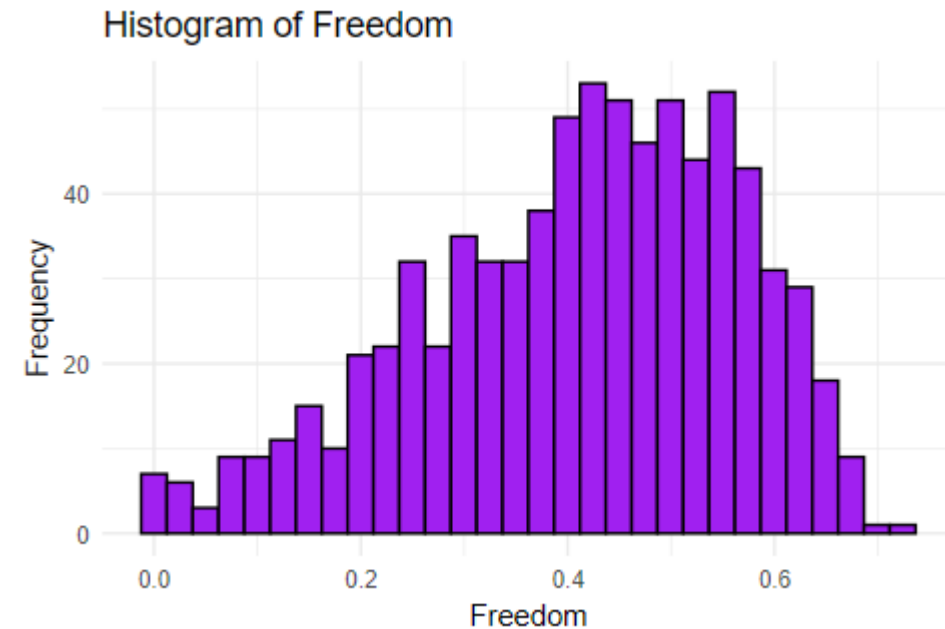
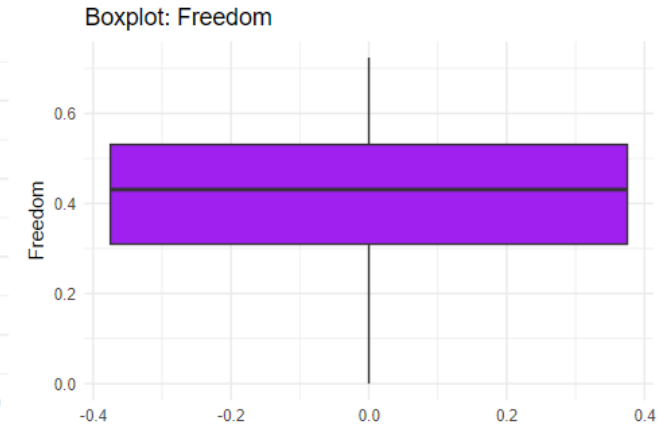
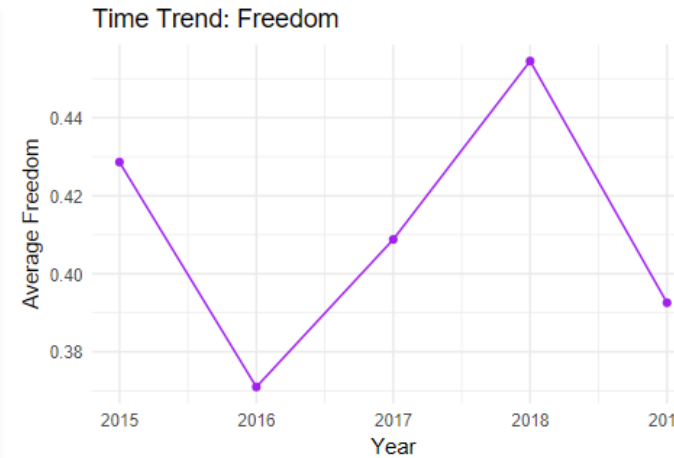
Healthy Life Expectancy

- A few slightly lower or higher outliers
- No major skew
- Healthy Life Expectancy likely won't need transformations



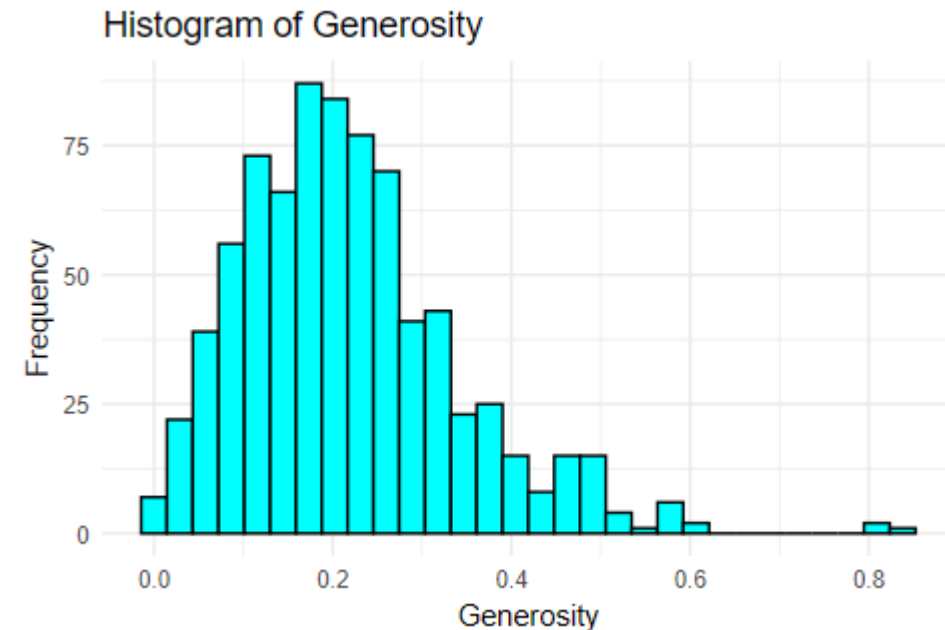
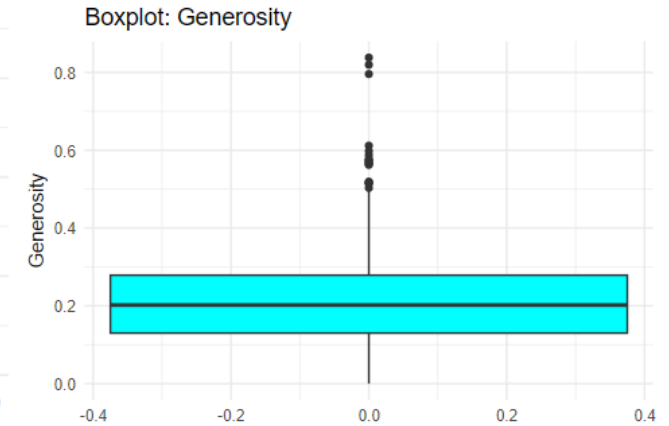
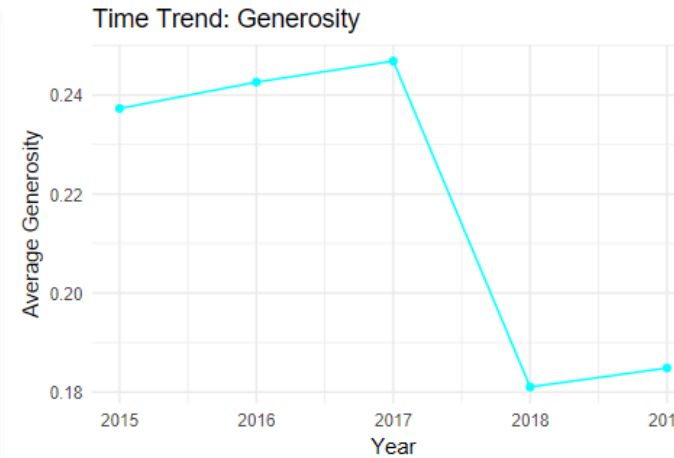
Freedom

- The median is fairly stable around 0.4-0.5
- fluctuation year to year
- A moderate peak near the center; no severe skew
- Freedom is a bounded variable (0-1). If any countries report extremely low (near 0) or high (above 0.7) values, it could noticeably affect cluster formation.



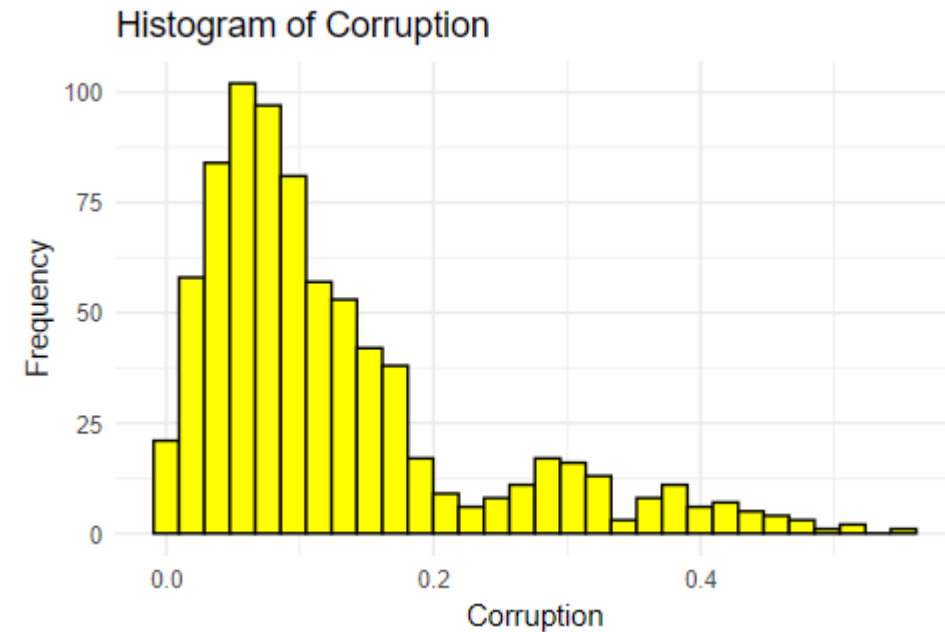
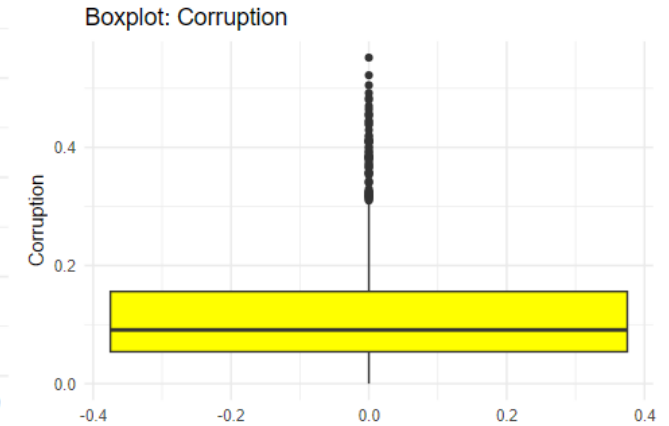
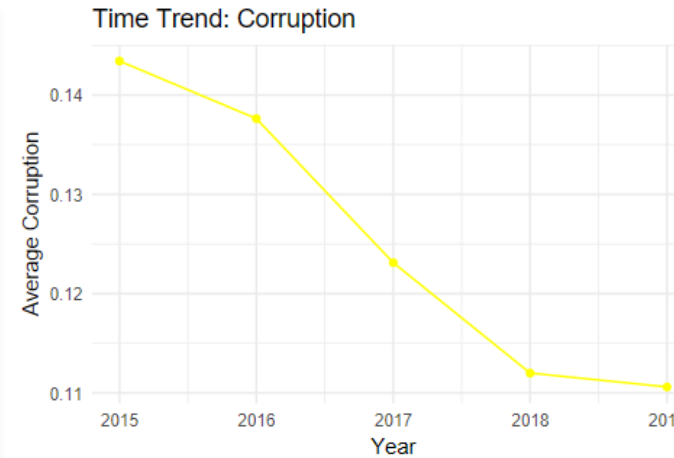
Generosity

- outliers above 0.6
- This could affect the clustering
- Highly right-skewed
- If a handful of countries are extremely high in generosity, we consider a transformation (like log or even a robust approach) to handle outliers before clustering.



Corruption

- Outliers at the upper end
- few countries are probably highly corrupted
- Because so many countries have low corruption scores and a small subset are relatively high, standardizing or applying a robust distance approach can prevent problems



Features Across Countries(Shiny Application)

Global Map of Happiness Indicators

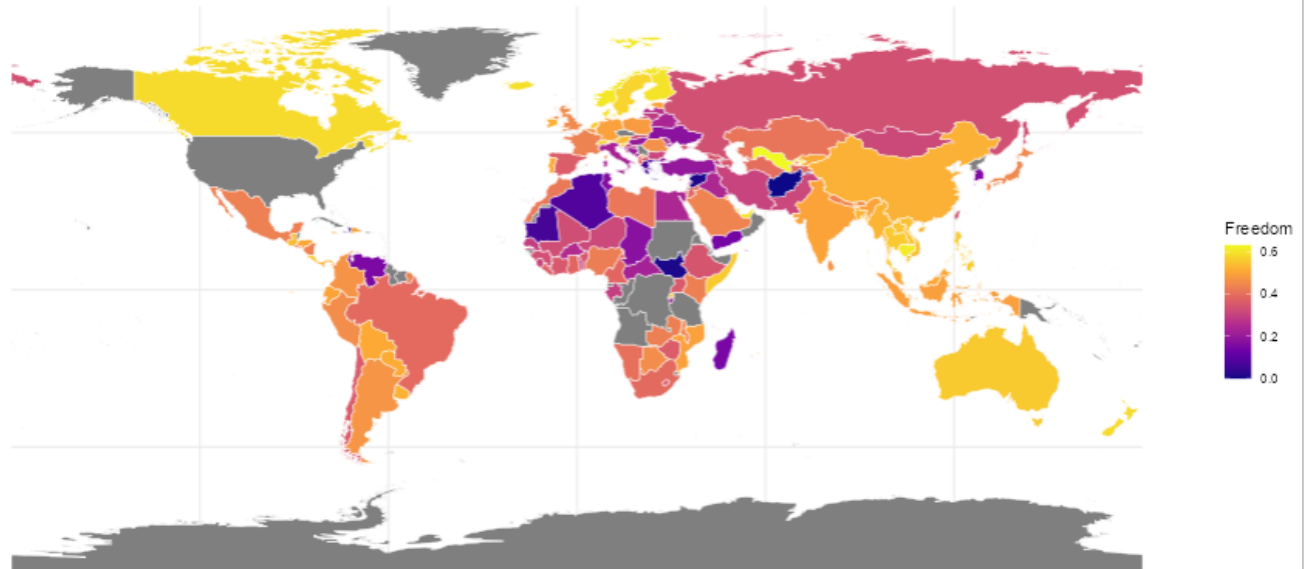
Select Feature:

Freedom

Select Year:

2019

Global Map of Freedom for 2019



Things to Consider

Transformations for Strongly Skewed Variables

- GDP per Capita
- Generosity
- Corruption

Potential Outlier Treatments

- extreme values in GDP or Generosity could severely pull cluster centers

Year-to-Year Variation

- Because many variables had fluctuations over time, It is better not to mix all of the data for clustering.
- It is more reasonable to perform a year-to-year analysis

Skewness

- Using the general rule **|skewness| > 1** to flag a “heavily skewed” distribution that might distort clustering
- Fortunately, None of the Variables were considered Heavily Skewed, So no transformation is needed

Variable	Skewness
Score	0.04
GDP per Capita	-0.32
Social Support	-0.68
Healthy Life Expectancy	-0.50
Freedom	-0.11
Generosity	0.30
Corruption	0.33

Outliers

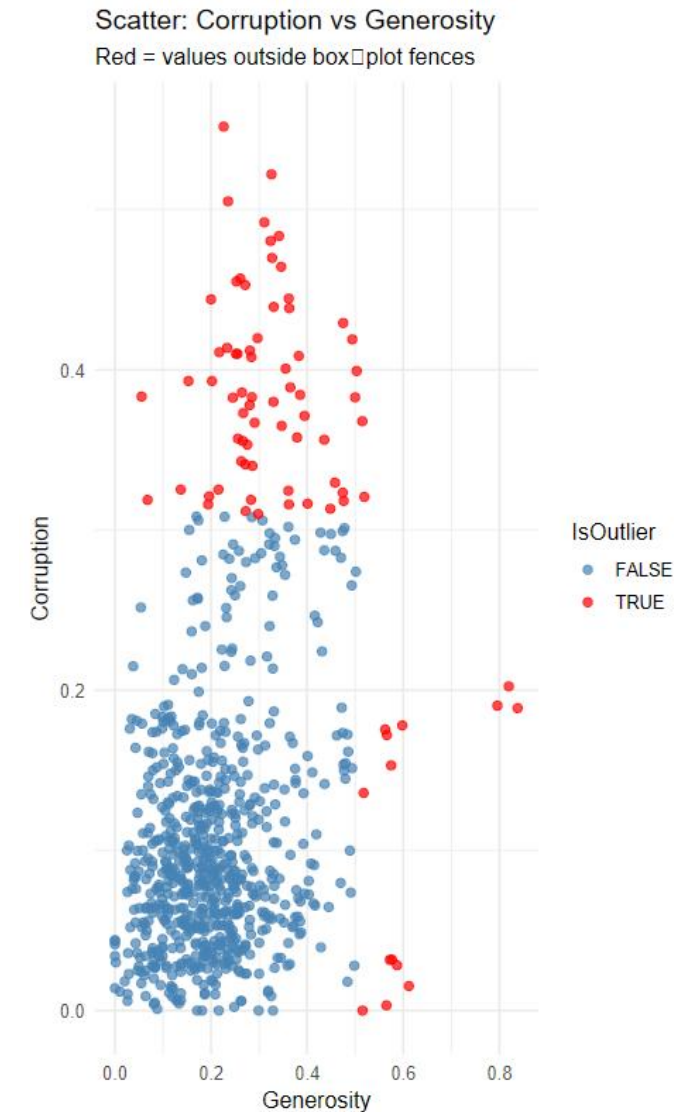
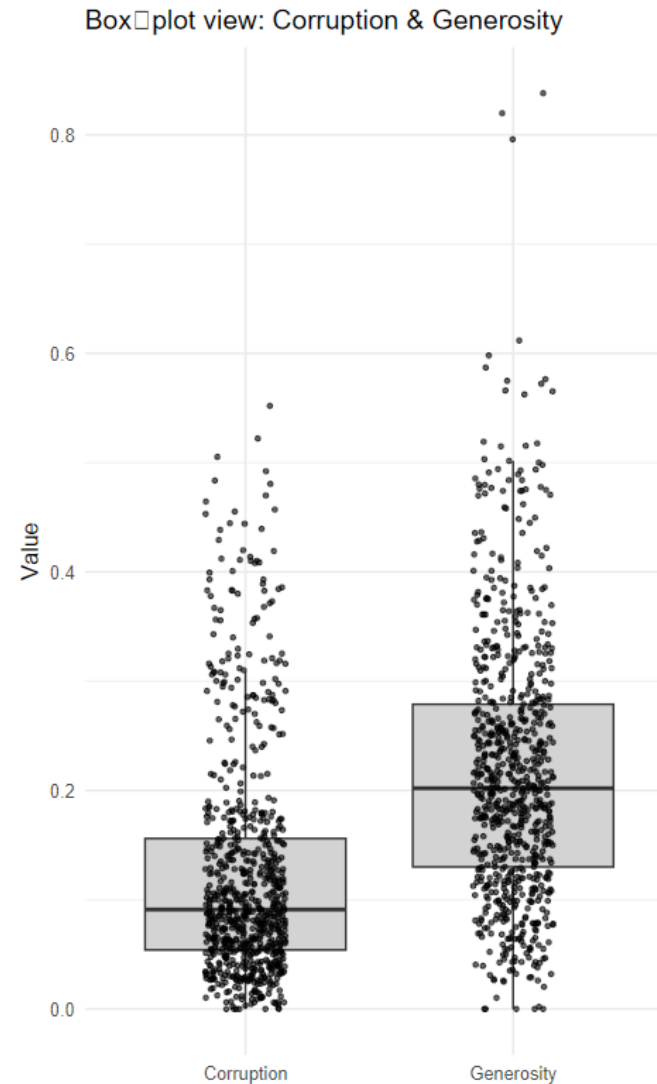
Variable	# of countries outside the box-plot fence	What kind of values?
Corruption	67	High scores (≥ 0.31)
Generosity	17	High scores (≥ 0.28)
All other variables	0	Nothing extreme under the $1.5 \times \text{IQR}$ rule.

Outliers

As we have observed before, There were outliers for the corruption and generosity features across the datasets.

Distance based algorithms give every variable equal weight. If one variable has many extreme values, it can dominate the distance measure and distort cluster shapes.

Since there are a lot of outliers(84), Instead of removing them which will result in valuable data loss, we could use **robust scaling (median / MAD)** for all the variables.



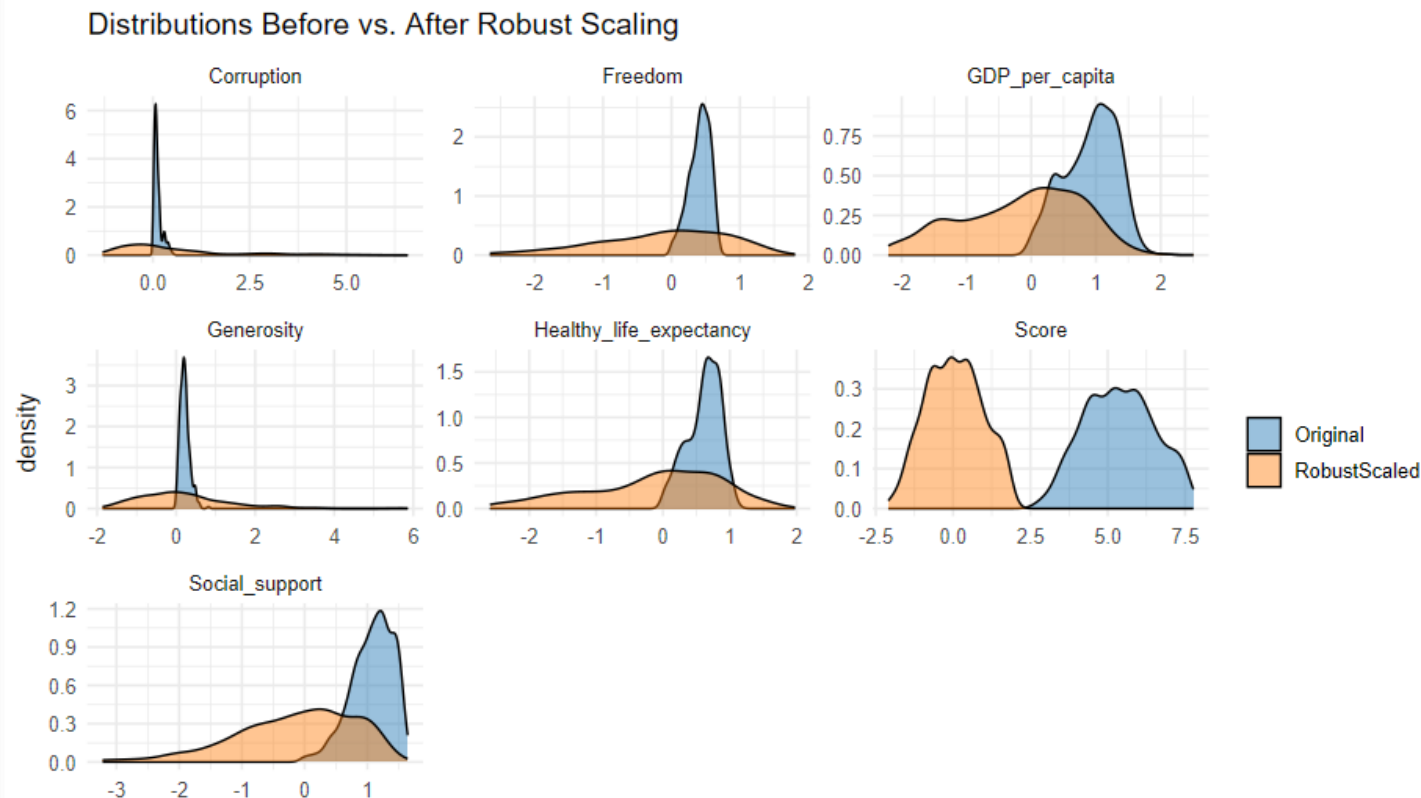
Robust Scaling

What do we observe?

- Orange curves are all centered around 0
- Widths of orange curves are similar - every variable has nearly the same spread ($\approx \pm 1$ MAD).

Why does this matter?

- No single variable dominates the distance calculation.
- Differences between countries are measured on a common scale.



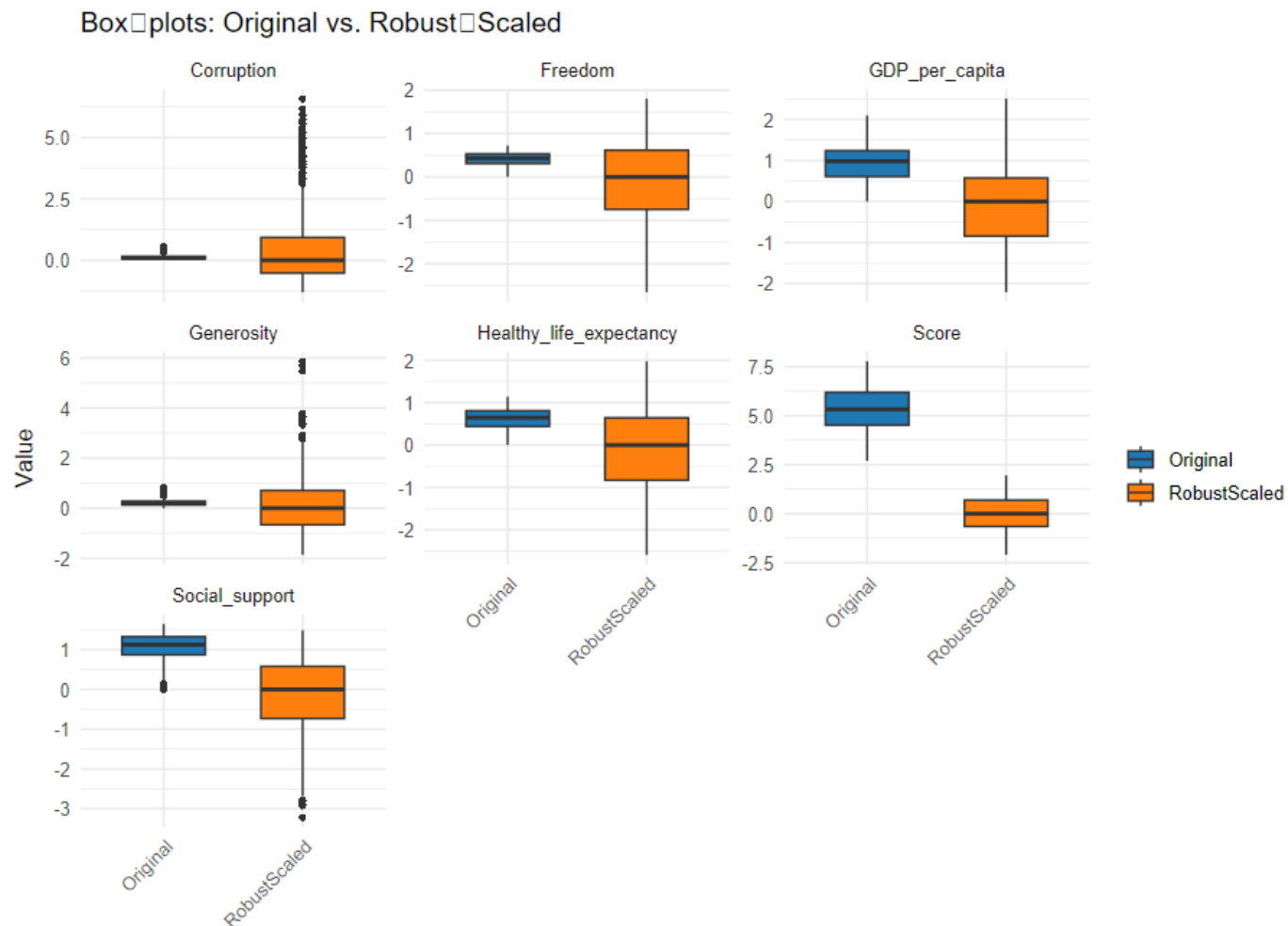
Robust Scaling

What do we observe?

- Outliers are still visible but less extreme on the new scale.
- Orange boxes all have the same median (0) and spread (≈ 1)

Why does this matter?

- Outliers keep their information but their numerical influence is reduced.
- Algorithms such as k-means, PAM, t-clust, PCA, etc., will treat each feature fairly



Robust distance matrix

- We already neutralized scale differences, now we must make sure “weird” countries don’t warp the shape of the cluster.
- A robust distance uses medians/MCD so a number of extremes cannot pull two ‘normal’ countries apart.
- Clustering algorithms judge “who is similar” purely from a **distance matrix**.
Raw Euclidean distance treats every column as perfectly clean and normally distributed → still sensitive to mild outliers and to different units if we later add mixed-type variables.
- Once the distance is in place, every downstream task becomes straightforward and consistent:
 - Dimensionality reduction (PCA/MDS, t-SNE, UMAP) can use the same distance.
 - Clustering methods (k-means, PAM, t-clust) all operate on that distance.
 - Validation and stability checks refer to the same geometry

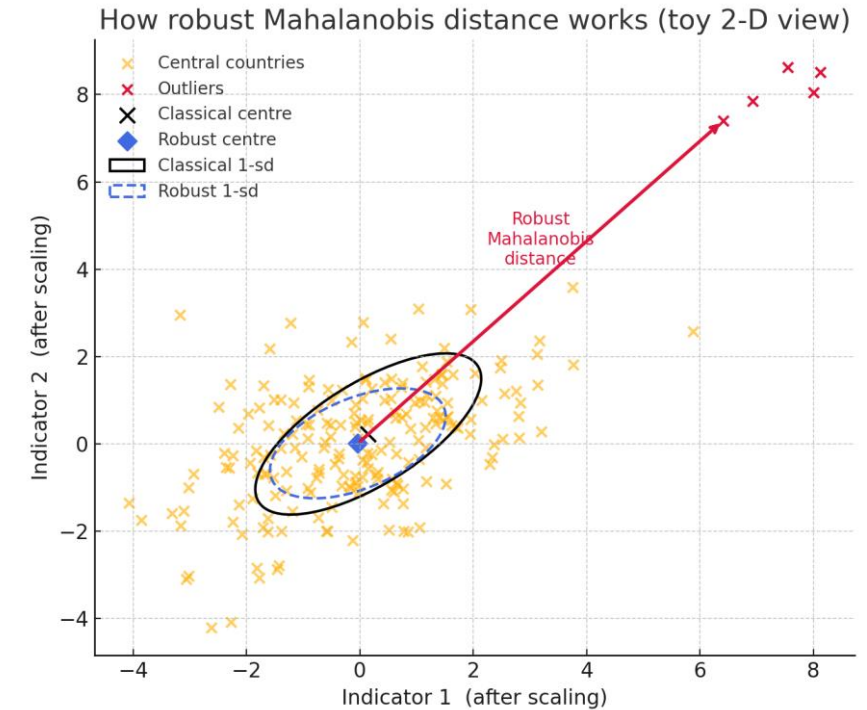
How far are countries from each other?

We need one simple number to say how similar or different two countries are before we can draw maps or form groups.

We can use **Robust Mahalanobis Distance**.
Why?

- Same scale for everyone
- Ignores double-counting
- Immune to outliers

It calculates the distance regarding the correlation in the dataset.



1. Robust centre (location):

$$\hat{\mu}_r = \text{vector of robust means}$$

2. Robust scatter (covariance):

$$\hat{\Sigma}_r = \text{robust covariance matrix}$$

3. Robust Mahalanobis distance for record x_i :

$$\text{RMD}_i = \sqrt{(x_i - \hat{\mu}_r)^\top \hat{\Sigma}_r^{-1} (x_i - \hat{\mu}_r)}$$

4. Pair-wise robust distance between two records x_i, x_j :

$$d_{ij}^{\text{robust}} = \left\| (x_i - \hat{\mu}_r) \hat{\Sigma}_r^{-1/2} - (x_j - \hat{\mu}_r) \hat{\Sigma}_r^{-1/2} \right\|_2$$

Multi-Dimensional Scaling (MDS)

Because we have 7 features, we cannot visualize the datapoints. Hence, we will use a method to shrink the 7 features into a 2-D space.

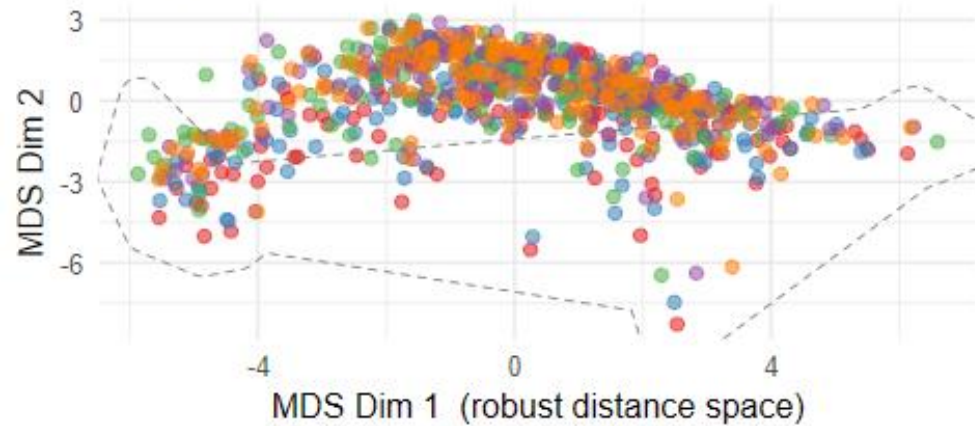
MDS is a way to turn a set of pairwise distances (here, between country-year records in 7-D) into points on a flat map (2-D) so that the map's distances match the originals as closely as possible.

Plots:

- Dots of different years sit almost on top of one another, so individual countries barely move their position in this 2-D space from 2015 to 2019
- Some countries are consistent outliers
- Robust Mahalanobis distances are capturing a clear, stable pattern(visible and consistent arch)
- Since the shape is stable across time, clusters will probably be meaningful and reliable

Classical MDS – Panel Evolution (2015–2019)

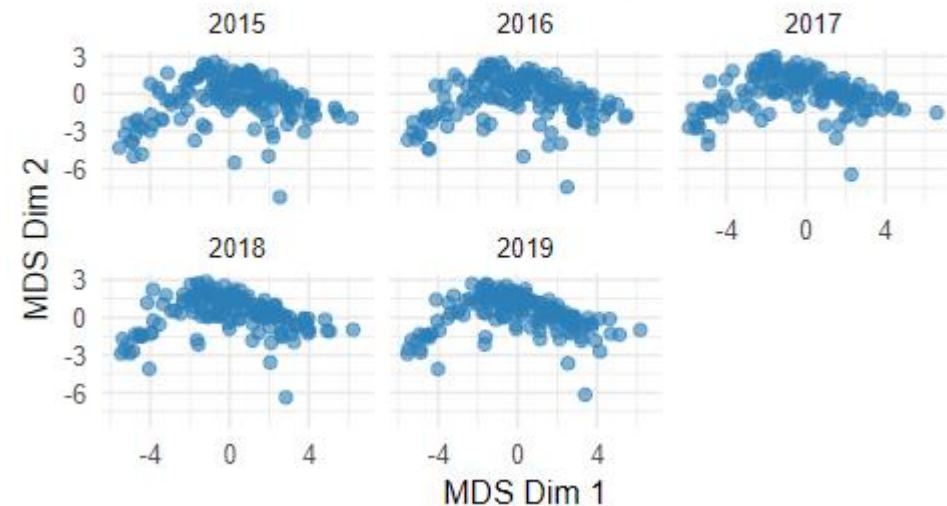
Every dot = a country-year record · dashed hull = top 5 % m



Year ● 2015 ● 2016 ● 2017 ● 2018 ● 2019

Classical MDS by Year (Robust Mahalanobis Distances)

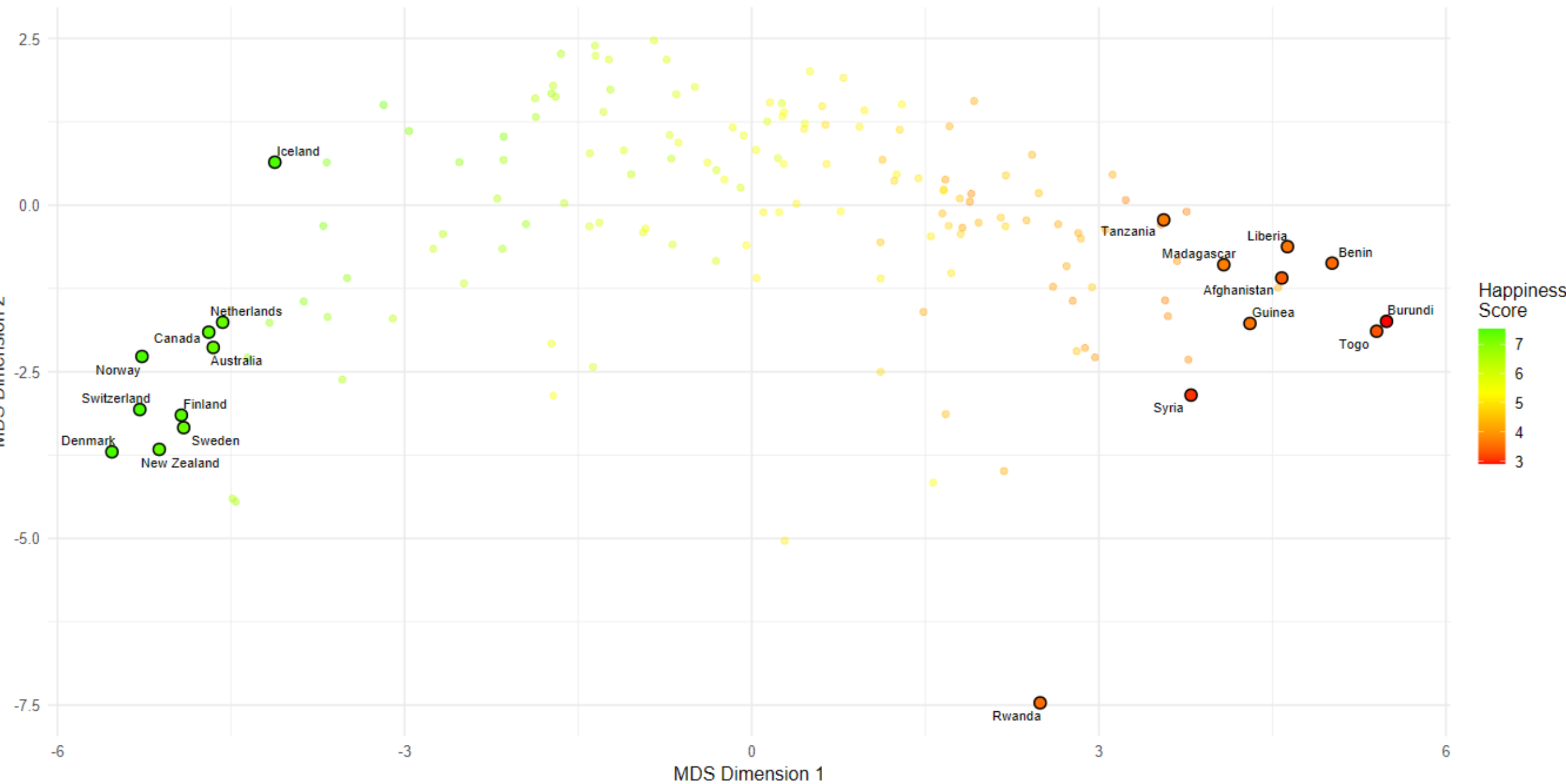
Dot = country position within its own year



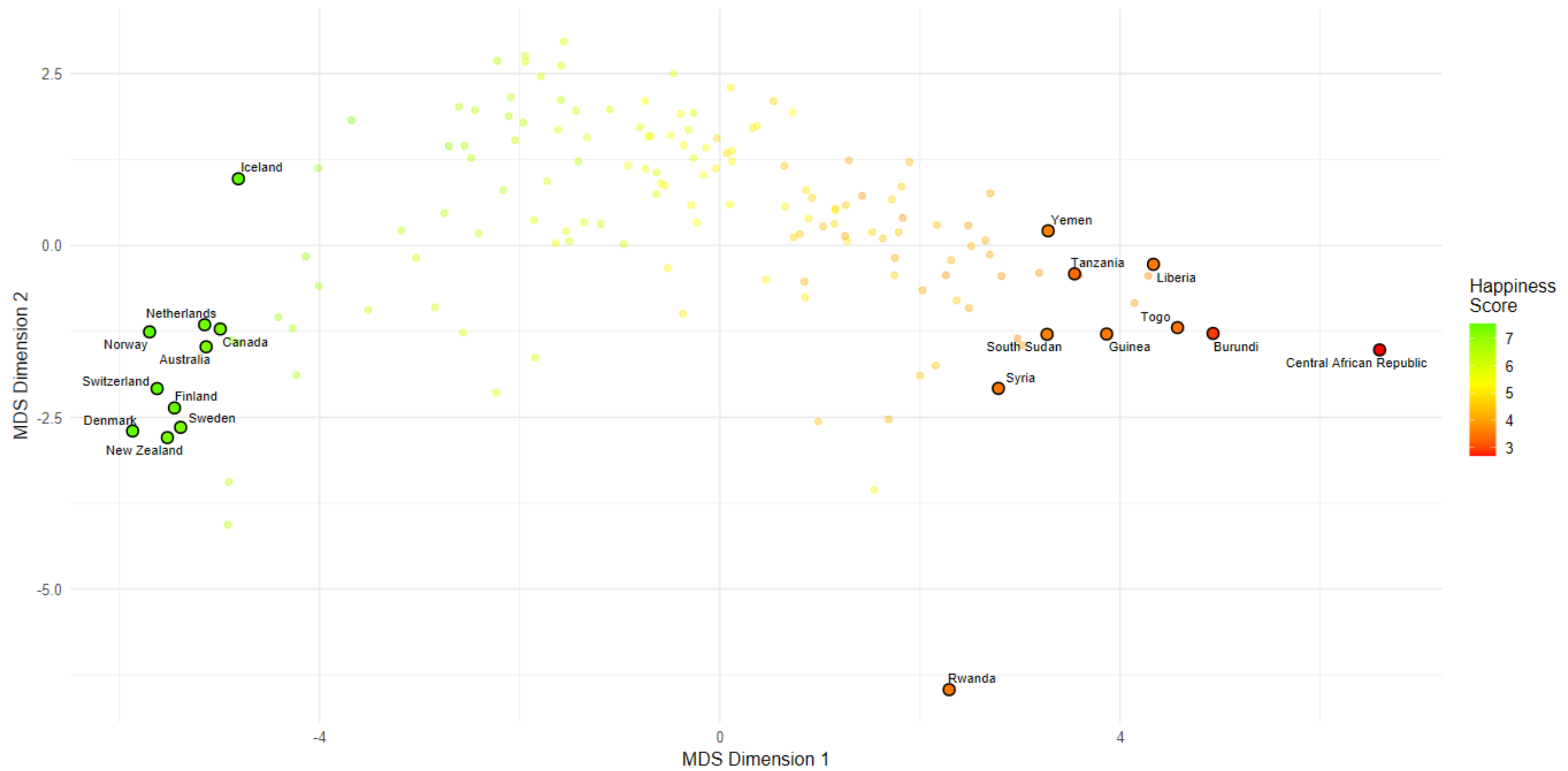
MDS Map for Year 2015



MDS Map for Year 2016



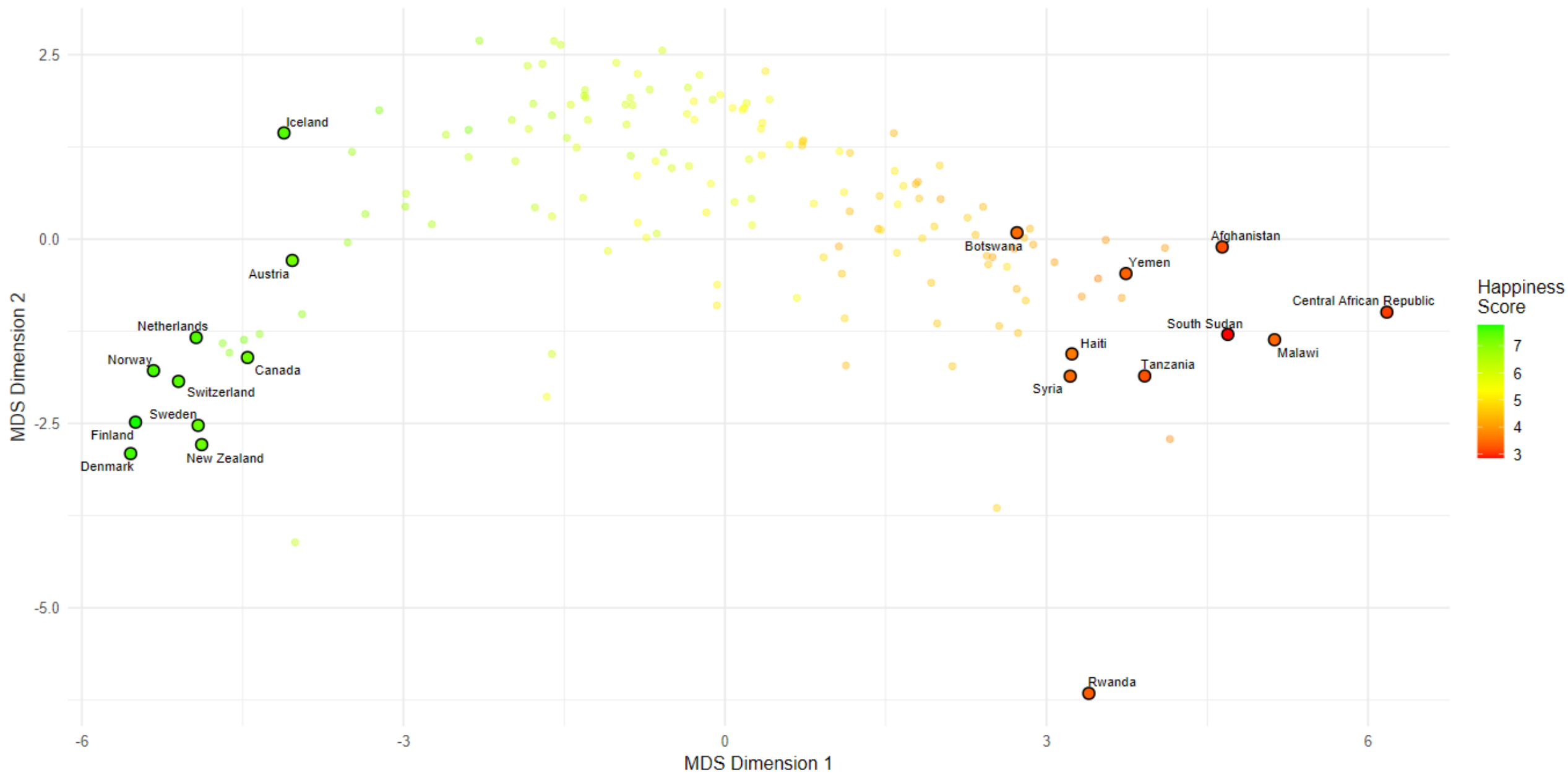
MDS Map for Year 2017



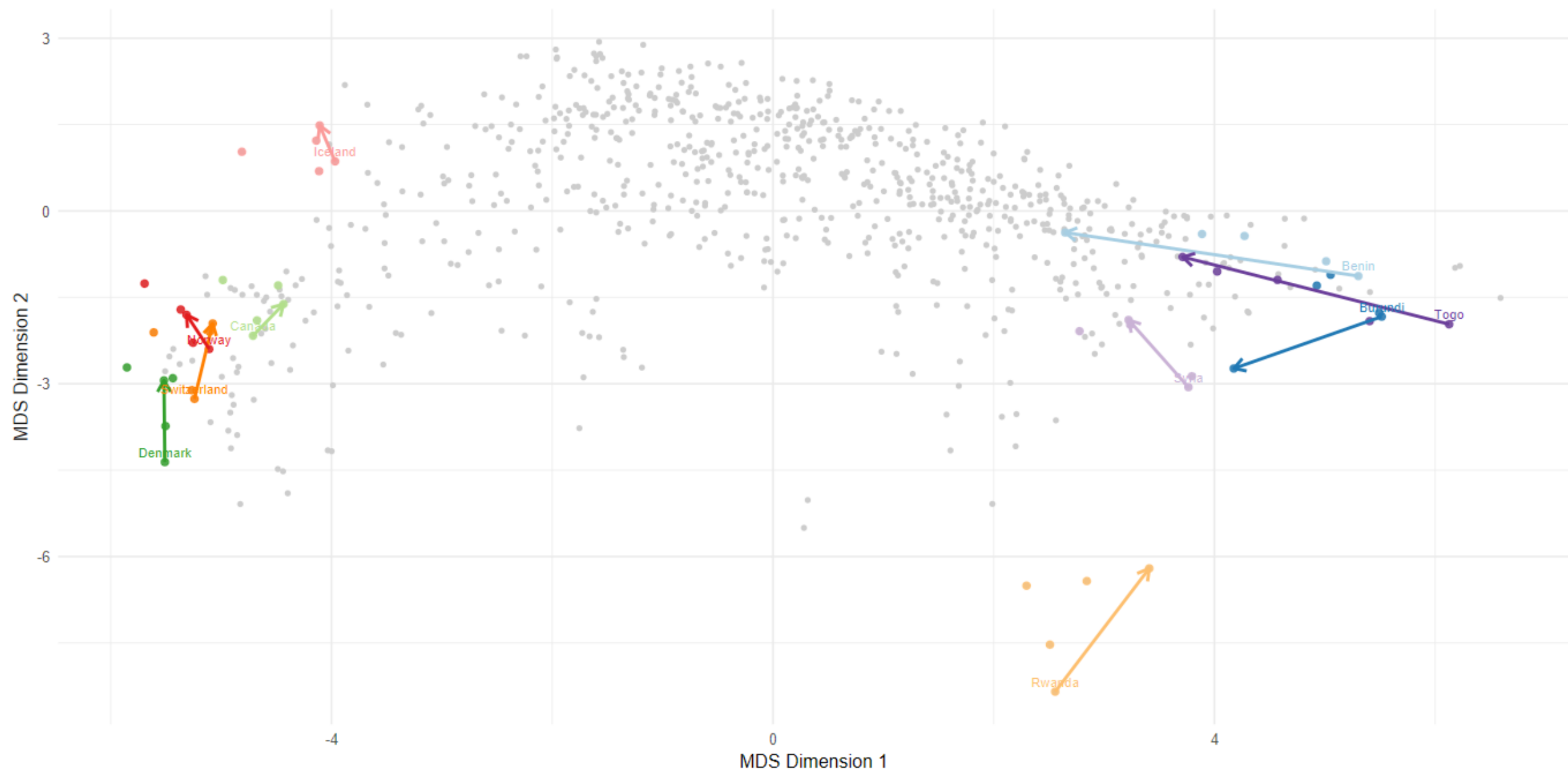
MDS Map for Year 2018



MDS Map for Year 2019



Country Movements on MDS: 2015 → 2019

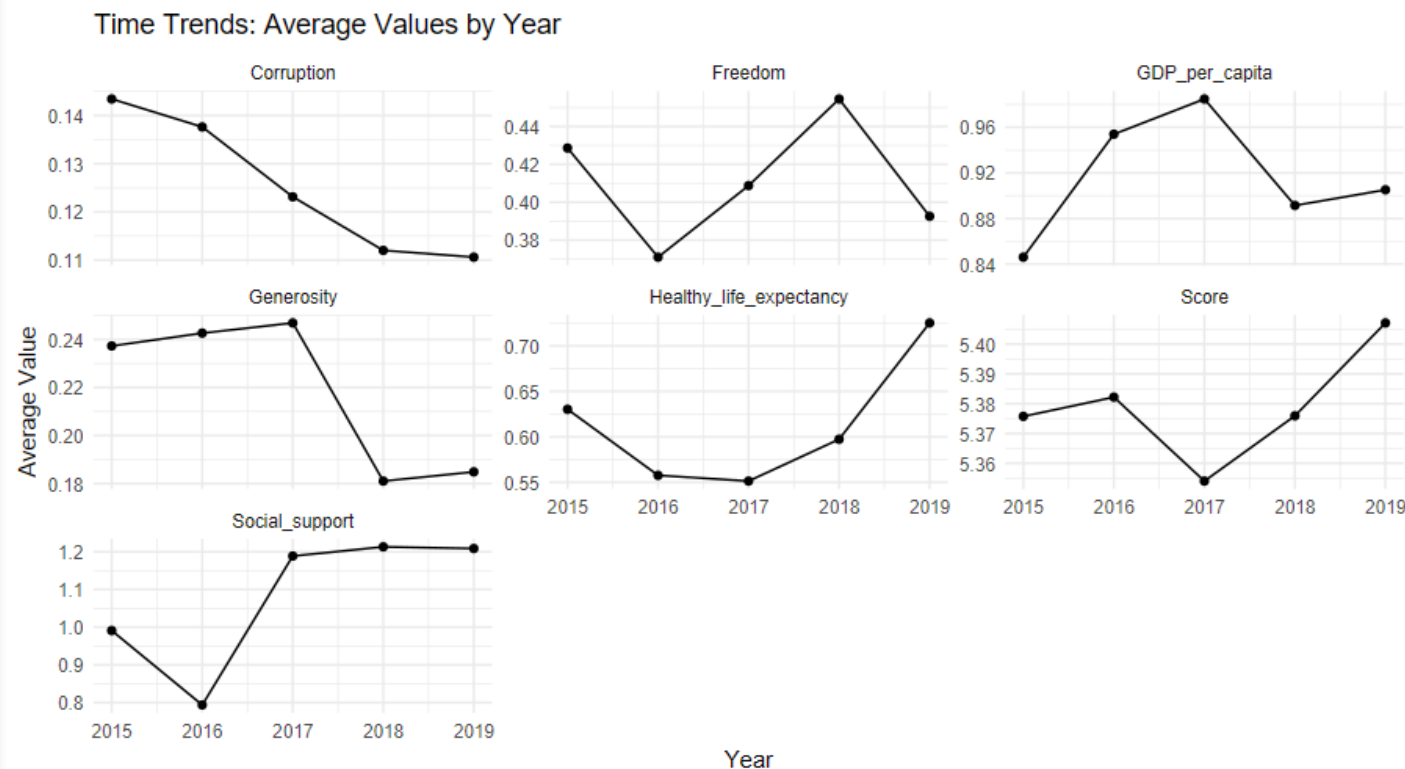


Features across time

As seen by the Time Trends, There are dips and spikes over the years for most of the features. These can cause a problem in segmentation between countries. If **too extreme**, it could cause problems for the clustering.

Time can trick clustering algorithms into grouping observations by year instead of by similarity, because:

- Distances get distorted when entire years shift up or down
- Measurement changes create incomparable scales

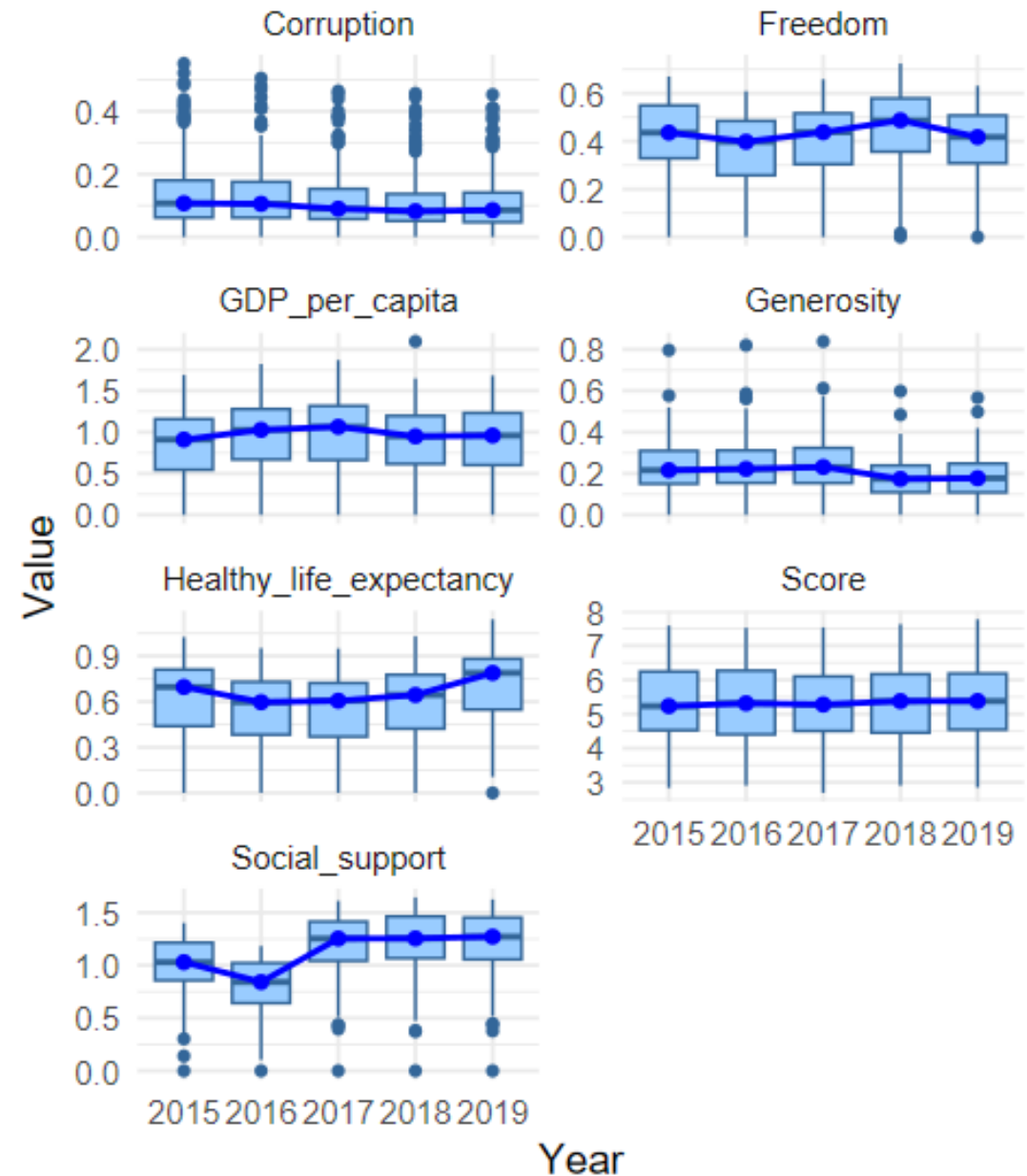


Features across time

From a closer and more technical point of view, the features over the year seem fairly stable.

So this probably is not going to make a problem for the clustering algorithms

Yearly Distributions by Feature (2015–2019)



Features Selection

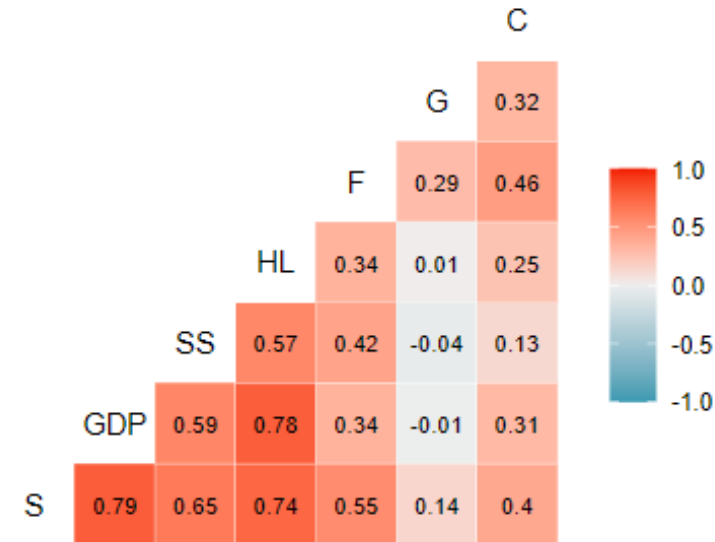
- *GDP per capita* and *Social support* are highly correlated so we only keep one of them
- *Score* is a weighted sum of all other factors and it doesn't add new dimensions of variation

Modification:

1. Removing "Social Support"
2. Removing "Score"

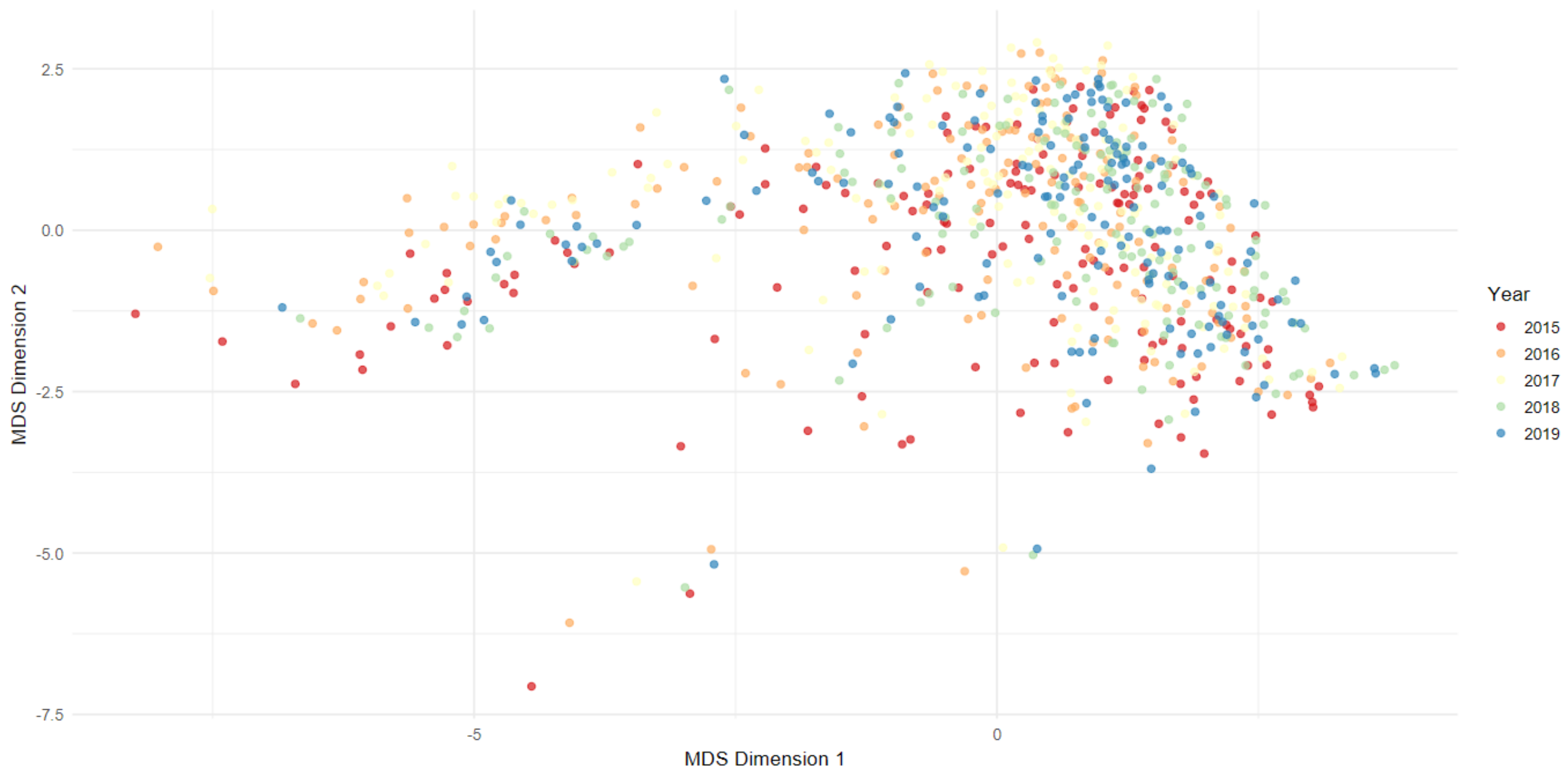
After this, we will check out the plots again.

Correlation Matrix (Abbreviated Variable Names)

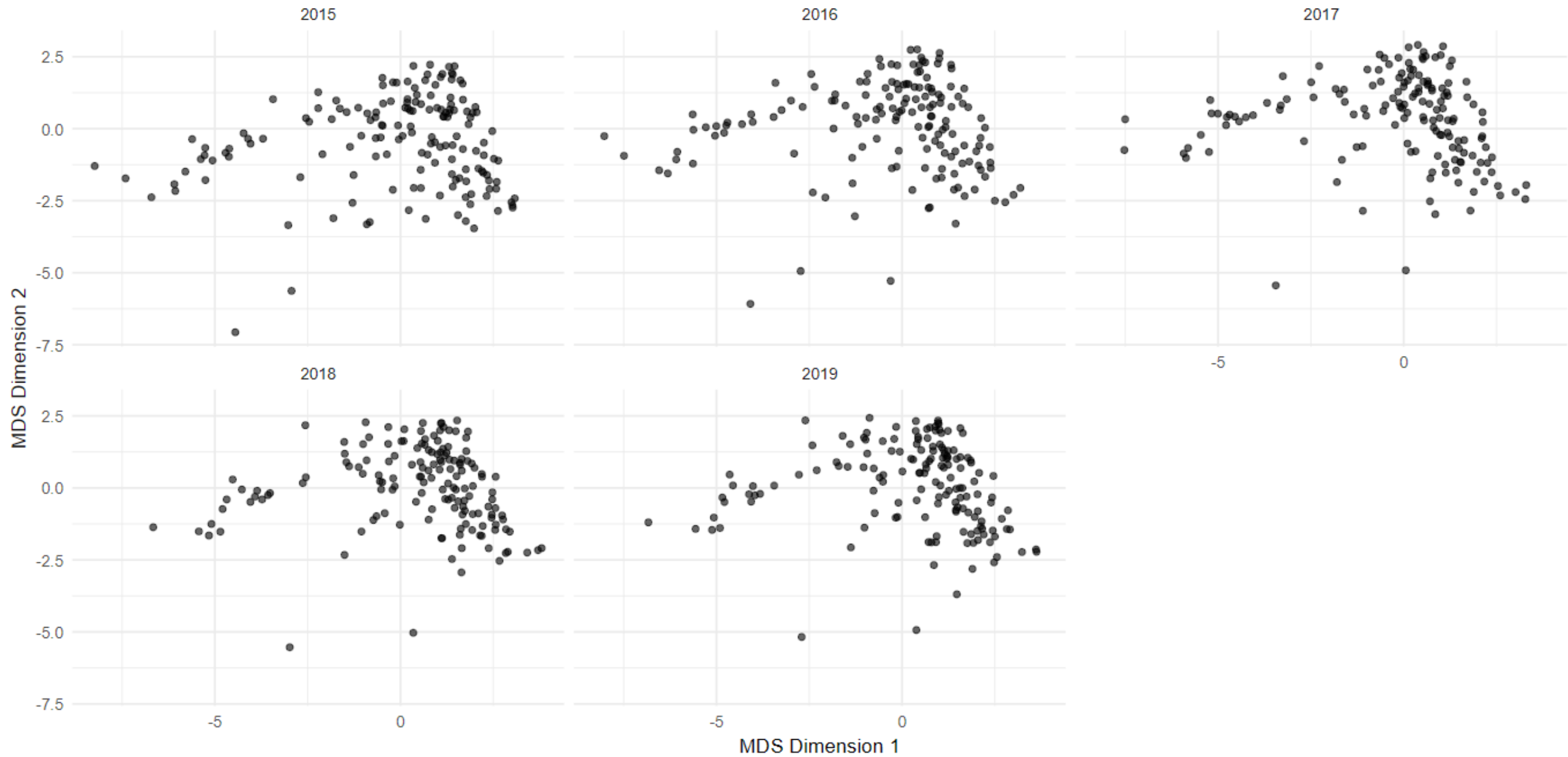


Pair	Correaltion	Why it matters
GDP ↔ Social_support	0.78	Very high: these two move almost in lock-step
Score ↔ GDP	0.79	Total happiness score is driven largely by GDP
Score ↔ Healthy_life_exp	0.74	Likewise, life expectancy explains much of Score
Freedom ↔ Corruption	0.46	Moderate: more freedom → less perceived corruption
Other pairs	≤ 0.42	Weak-to-moderate, so each adds distinct info

MDS Map (2015–2019) on Selected Features

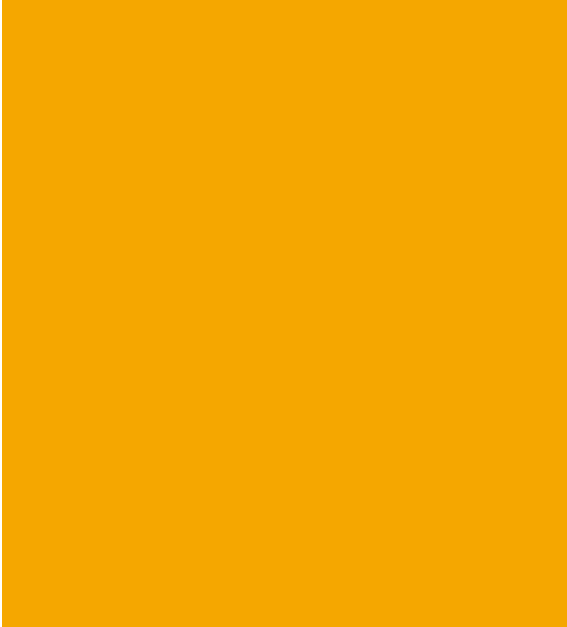


MDS Over Time (2015–2019) on Selected Features





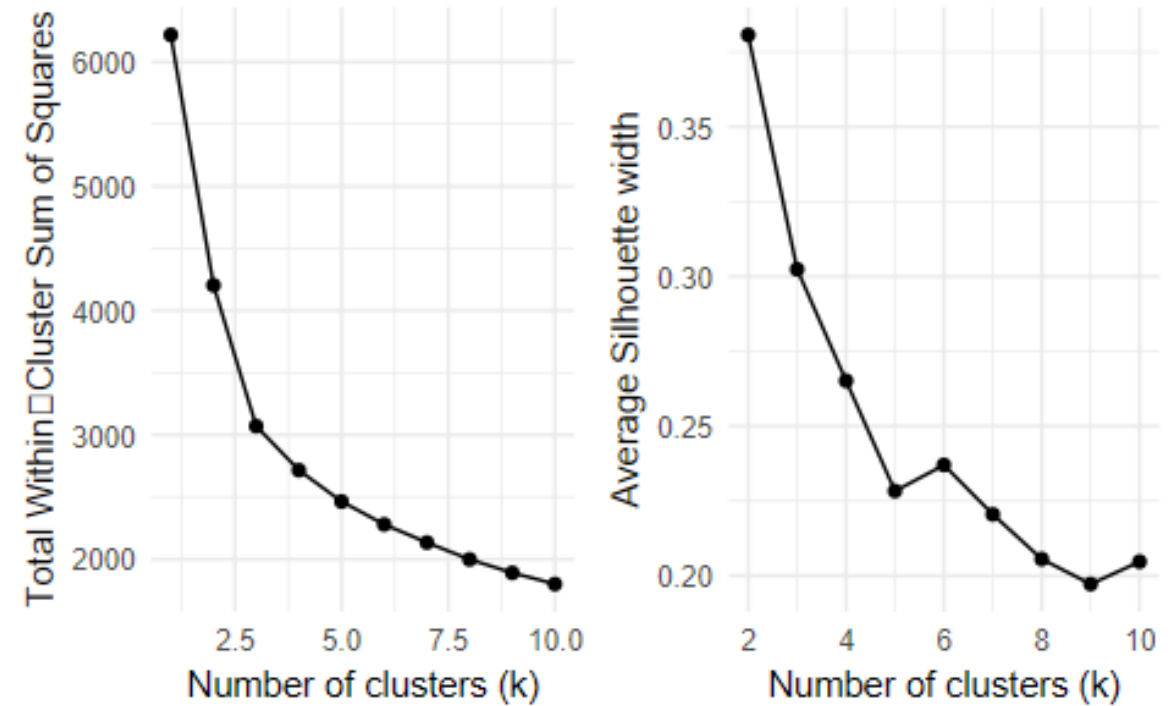
Models



K-means

- Elbow: $K=3$
- Silhouette: $K=2,3$
- 3-Cluster corresponding to “Low-Medium-High” happiness seems possible
- Now, we will apply the algorithm with $K=3$

Elbow Plot: k vs. Total WCSS Silhouette Scores: k vs.



MDS Map Over Time, Colored by K=3 Clusters

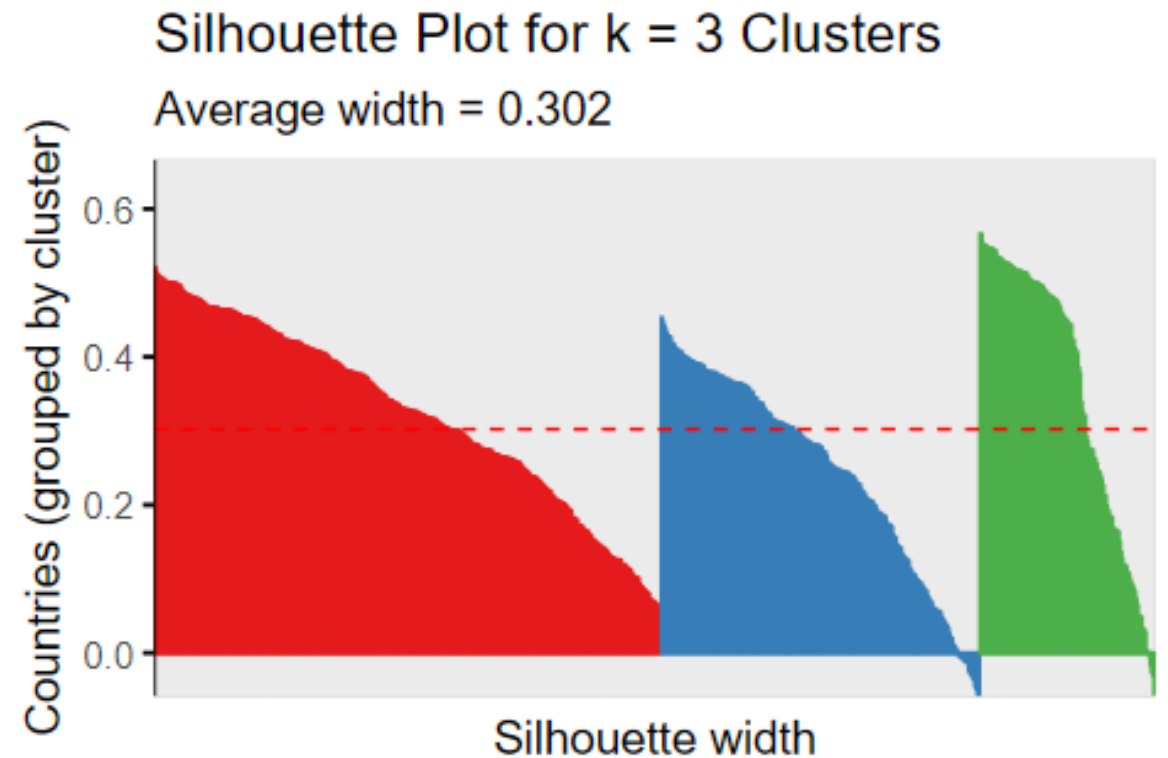


K-means(Cluster Profiles)

Cluster	Size	Avg. silhouette
1	396	0.32
2	249	0.24
3	137	0.35

Rules of thumb (Rousseeuw, 1987):

- More than 0.5 $\Rightarrow$ strong structure
- Between 0.25-0.50 $\Rightarrow$ reasonable structure
- Less than 0.25 $\Rightarrow$ no substantial structure



K-means(Cluster Profiles)

Cluster	Size	Avg Score	Avg GDP	Avg Support	Avg Life Exp	Avg Freedom	Avg Generosity	Avg Corruption
1	50%	5.65	1.05	1.19	0.67	0.41	0.22	0.13
2	39%	4.23	0.48	0.77	0.56	0.37	0.24	0.14
3	11%	6.70	1.33	1.30	0.91	0.52	0.43	0.30

K-means(Cluster Profiles)

Cluster 1: "Mid-Happiness"

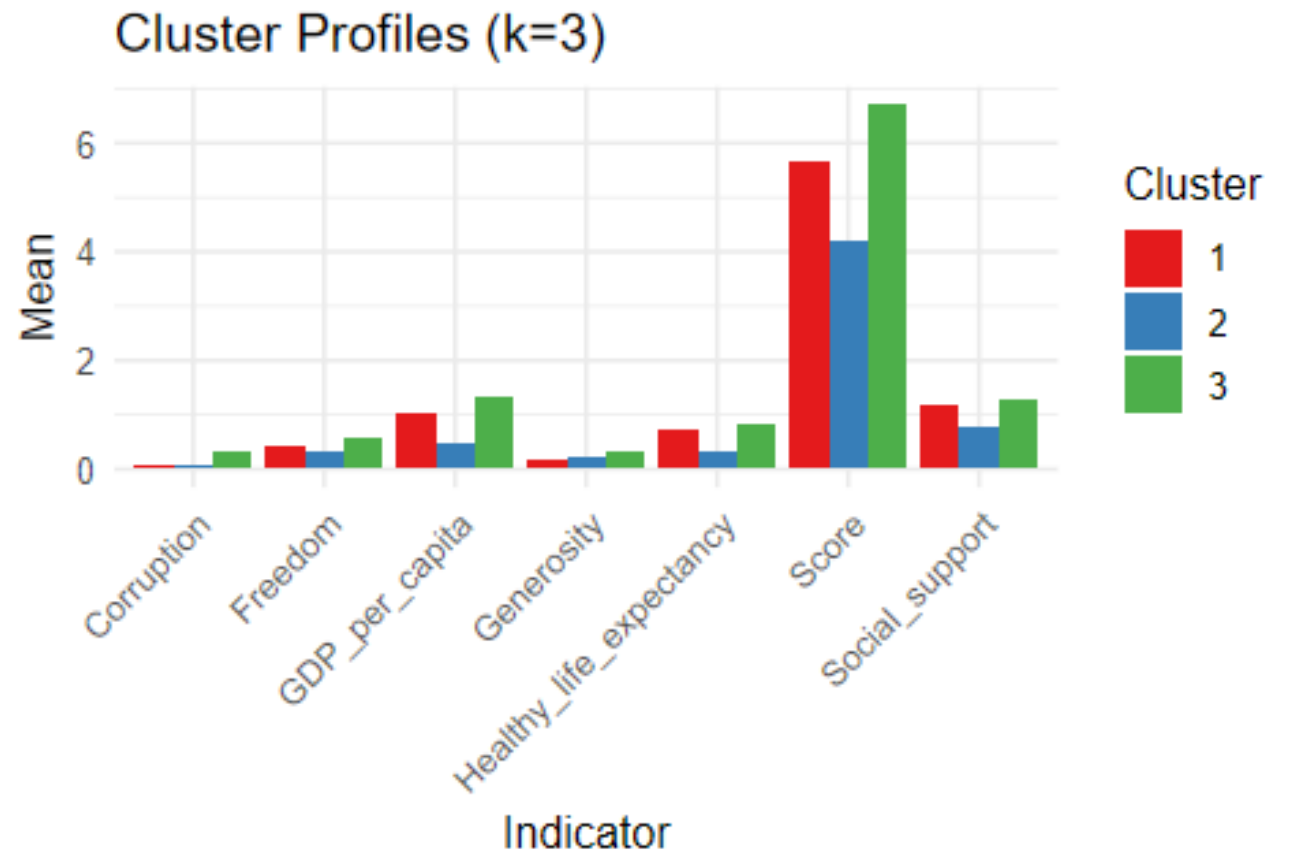
- GDP and social support near average
- Health & freedom around typical levels
- Below-average generosity & corruption perception

Cluster 2: "Low-Happiness"

- Markedly low GDP, support, health & freedom
- Generosity & corruption perception close to overall mean

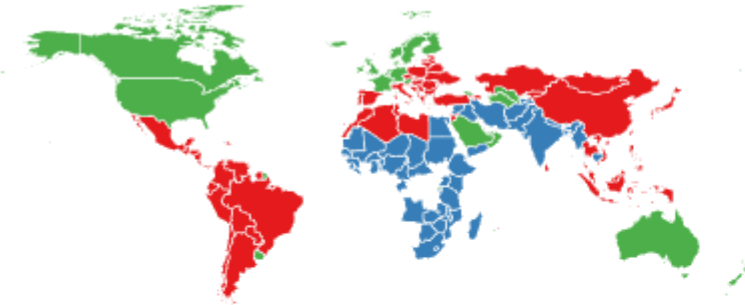
Cluster 3: "High-Happiness"

- Strong GDP, social support, health & freedom
- Above-average generosity & higher corruption

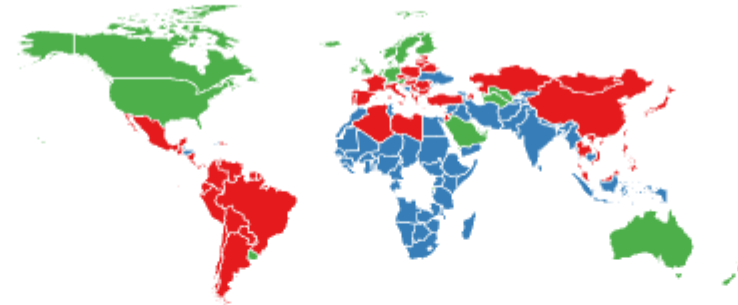


K-means(Clusters Over Time)

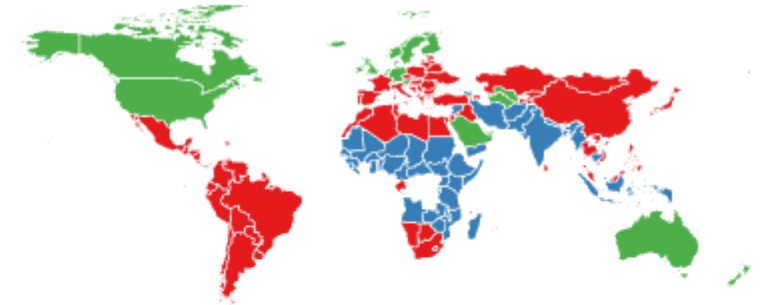
2015



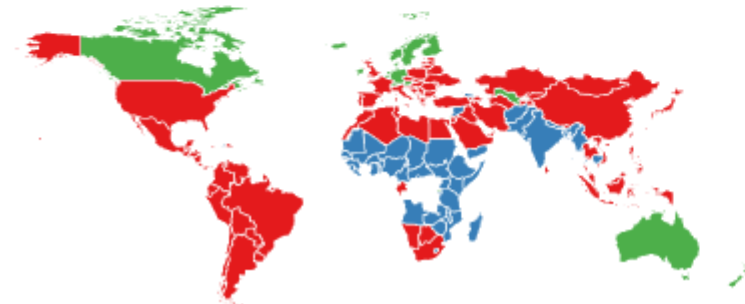
2016



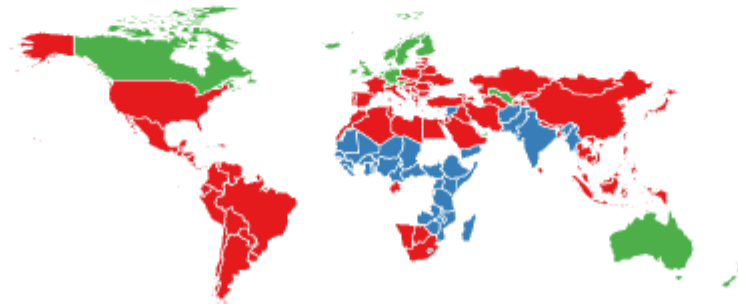
2017



2018



2019

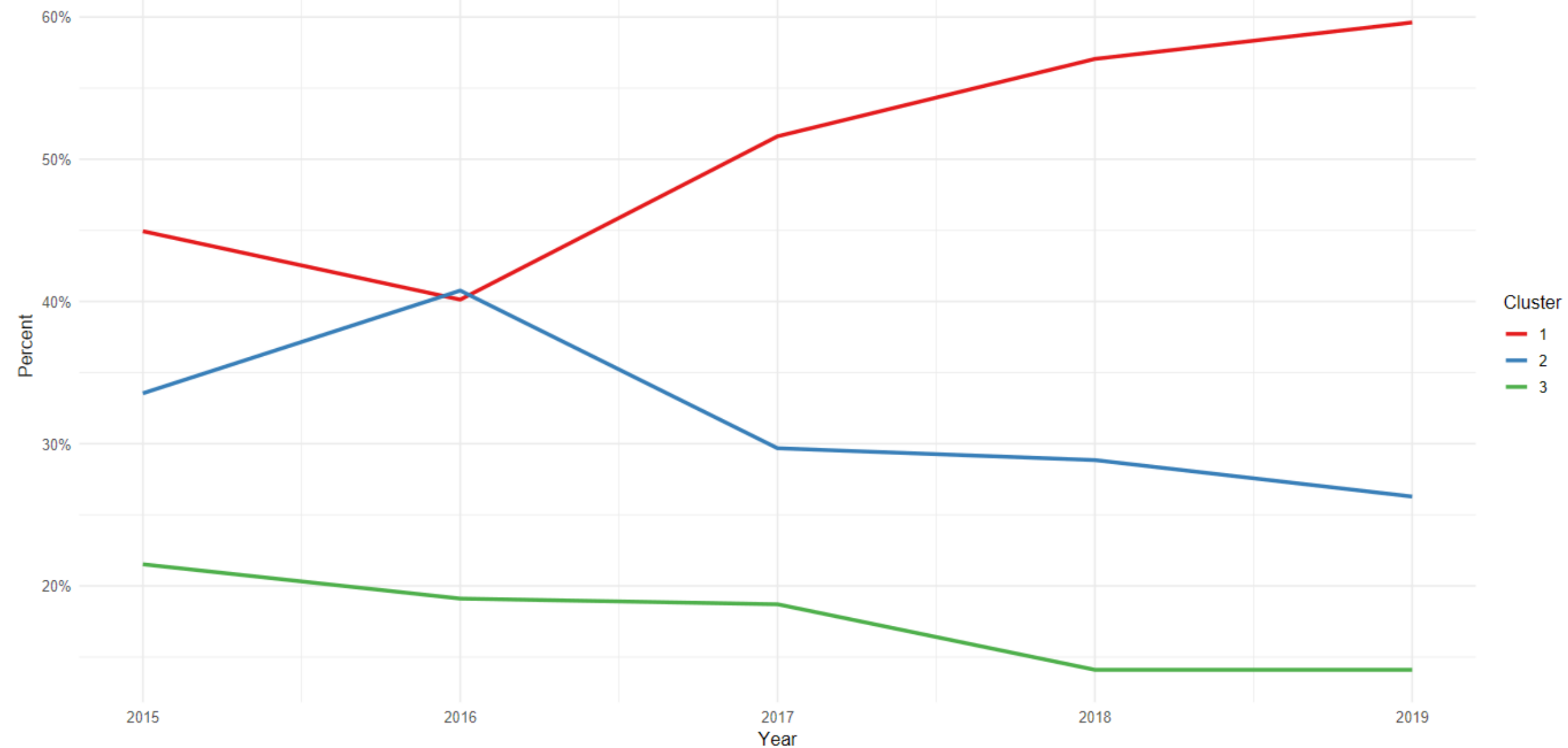


Cluster 1: "Mid-Happiness"

Cluster 2: "Low-Happiness"

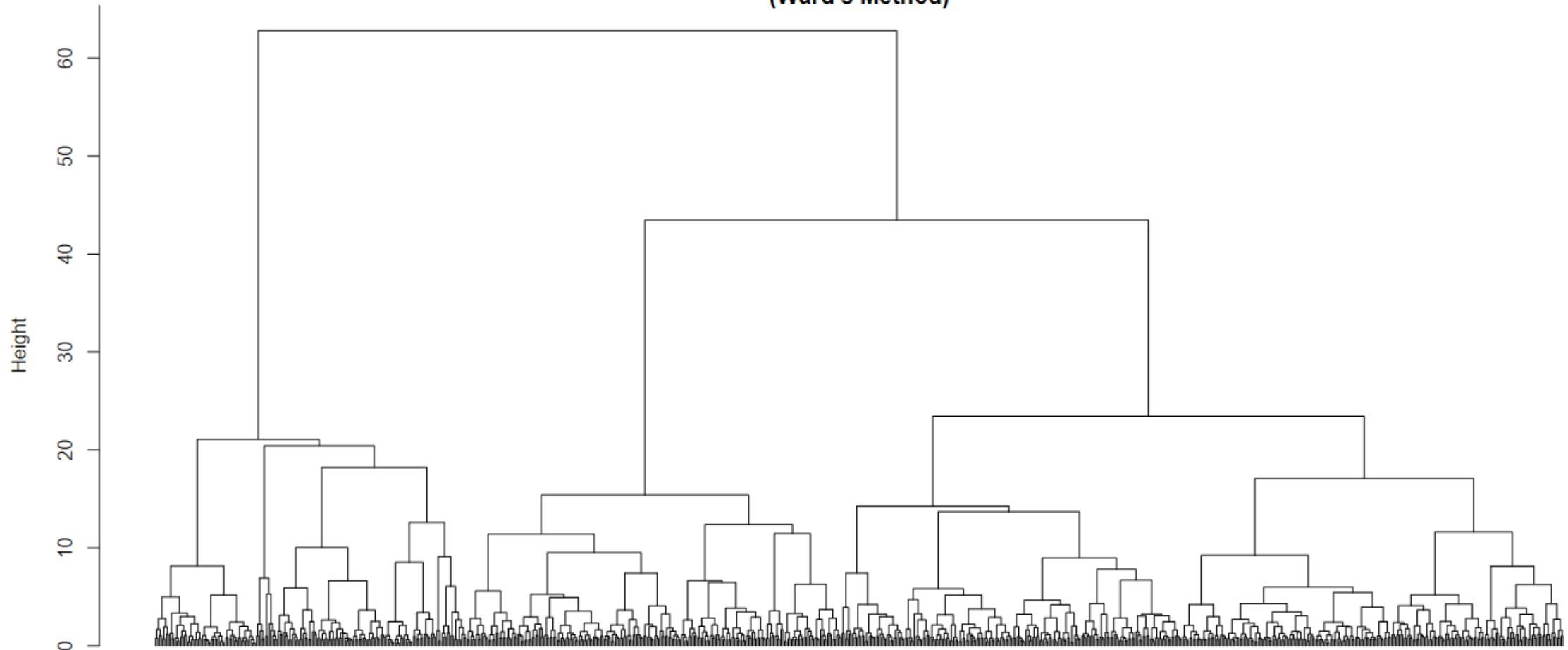
Cluster 3: "High-Happiness"

Proportion of Country-Years by Cluster (2015–2019)

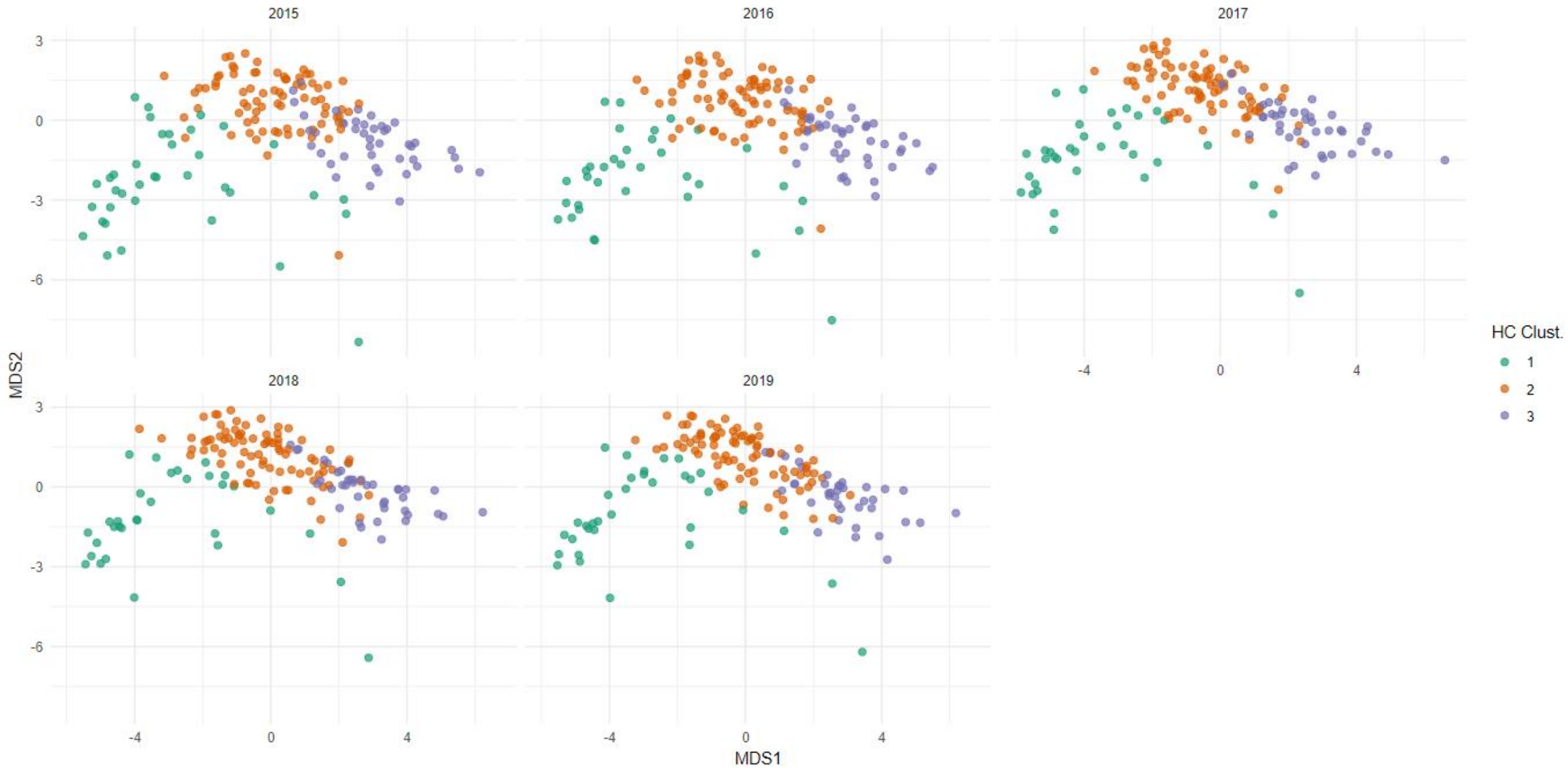


Hierarchical Clustering

**Hierarchical Clustering Dendrogram
(Ward's Method)**

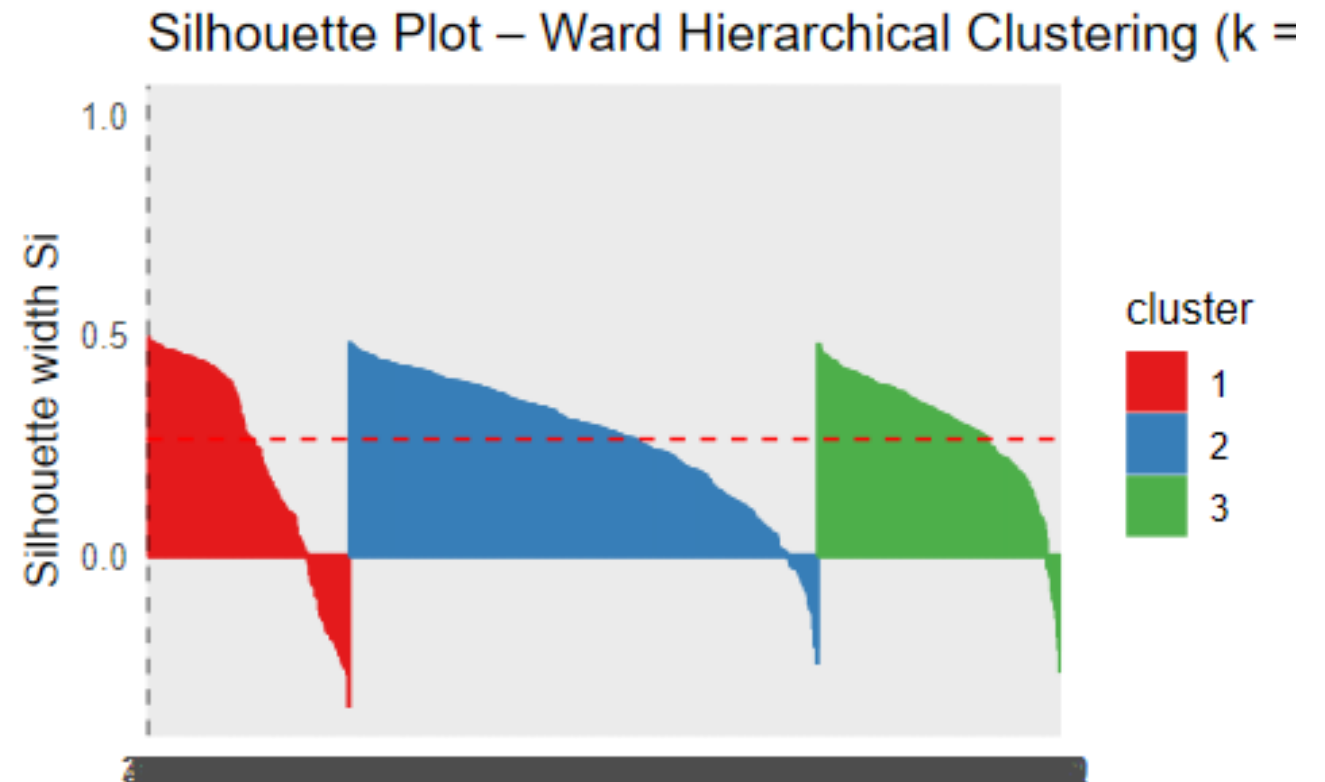


MDS Map – Hierarchical Clusters (k = 3)

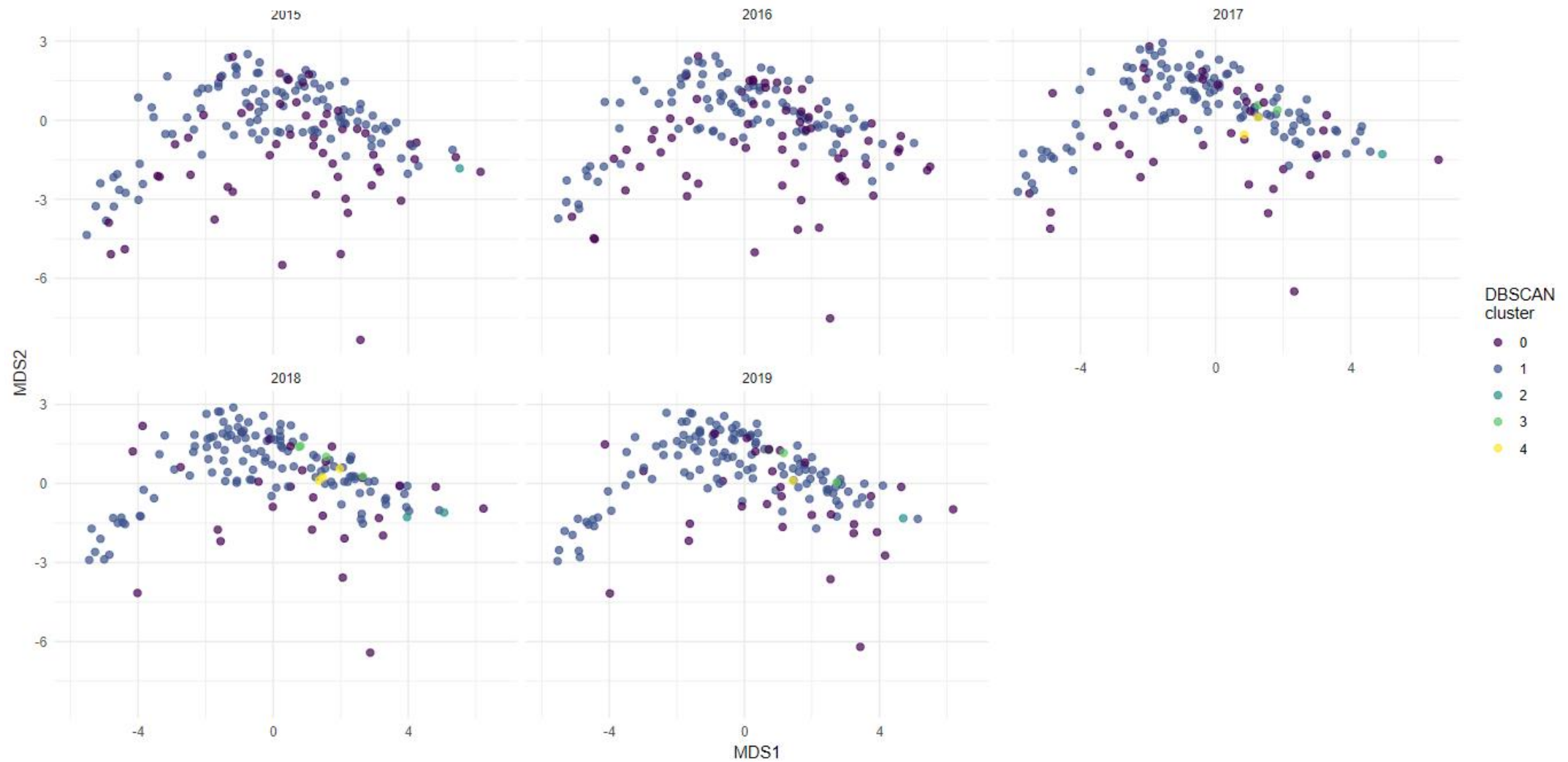


Hierarchical Clustering

- Avg = 0.264, lower than k-means.
- Data points are not so tight
- Clusters 2 & 3 are tighter (avg 0.27 and 0.29).

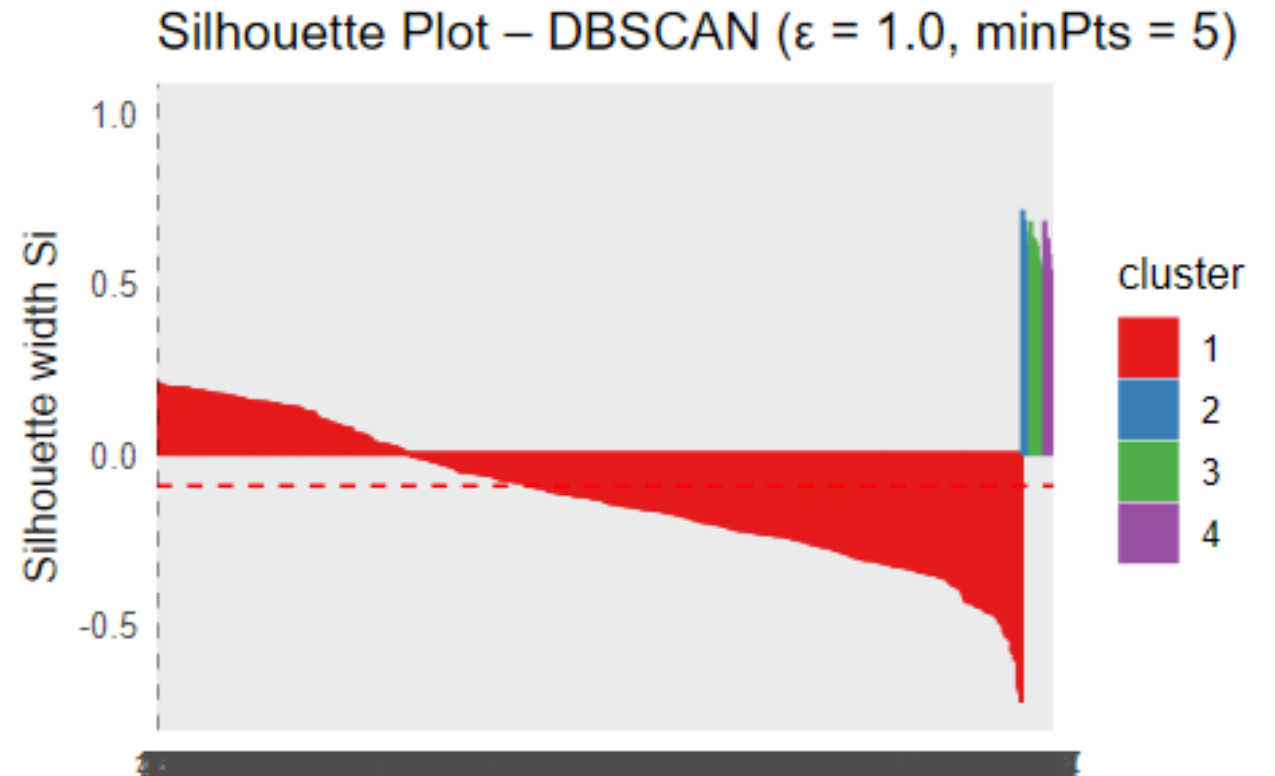


DBSCAN



DBSCAN

- Avg = -0.094 for the big cluster
- Most core points actually sit closer to points in the next cluster than to their own, signaling DBSCAN's single massive cluster is a poor fit
- The three small clusters have high silhouette widths but they're so small that they aren't very useful for broad segmentation





Additional Work



t-SNE
UMAP

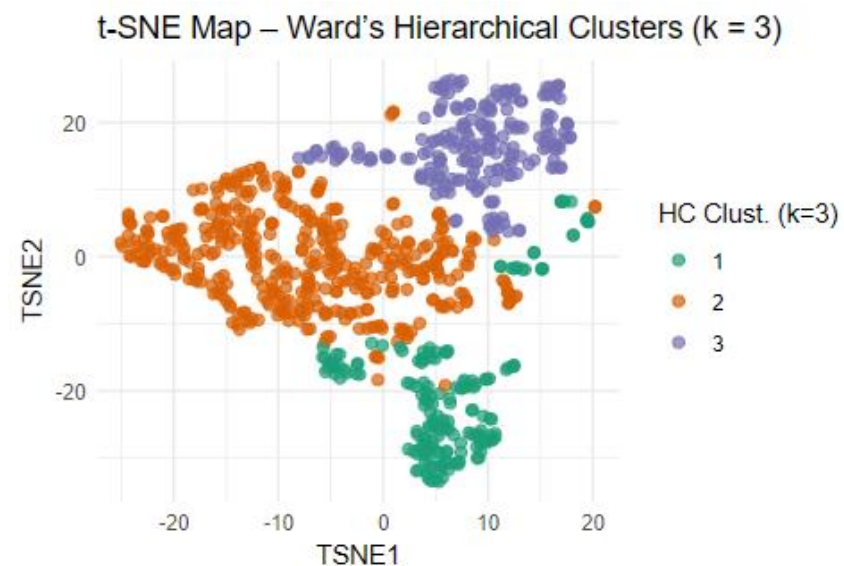
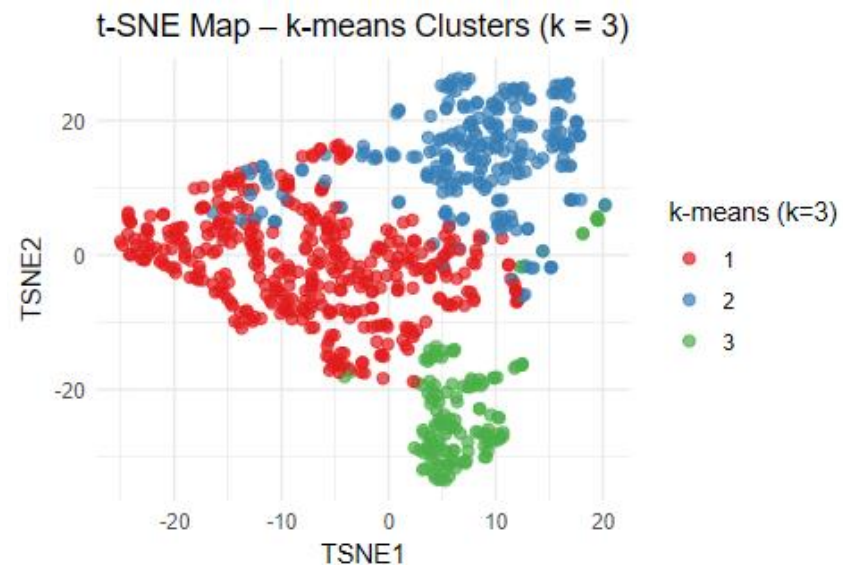
t-SNE

K-means:

- The clear gaps in t-SNE space show these clusters are reasonably well separated in the original 7-D feature space.

Hierarchical:

- Good coherence of the purple group and a orange mass.



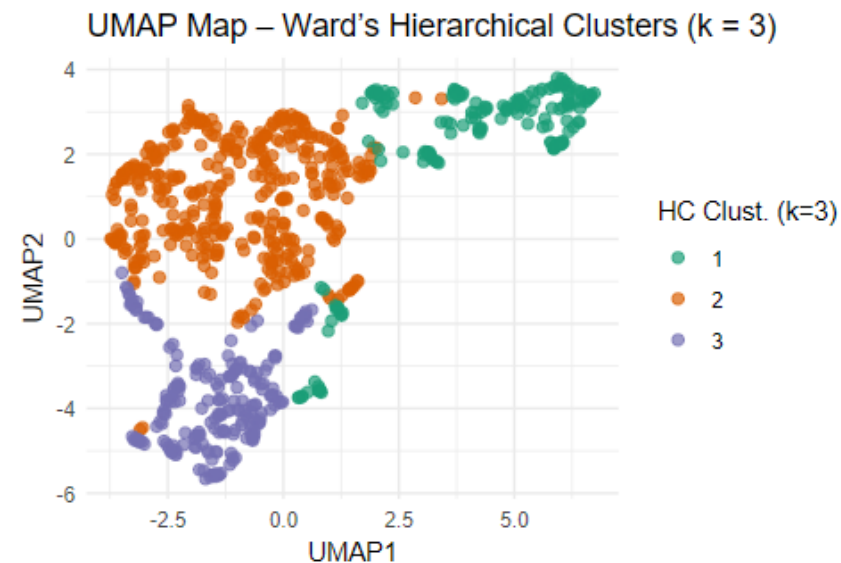
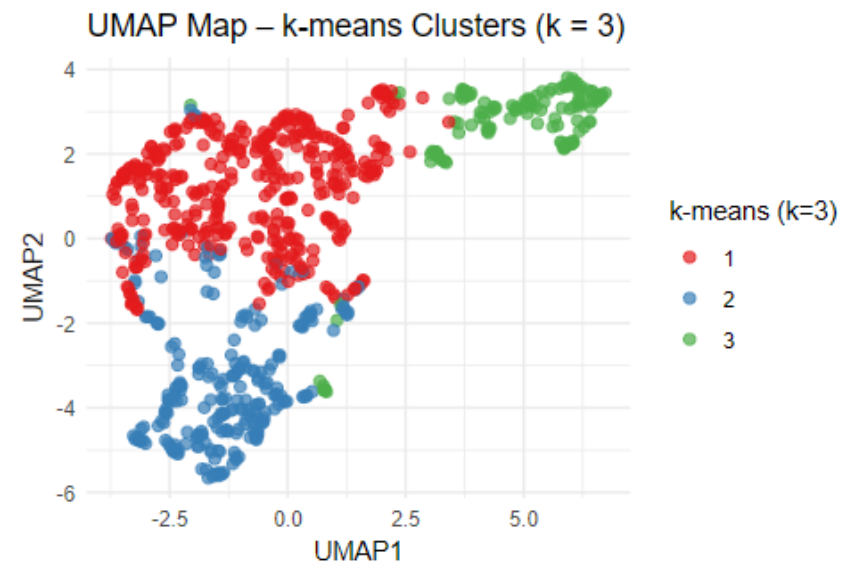
UMAP

K-means:

- Red cluster might actually hide two sub-patterns that t-SNE didn't resolve (just a possibility)

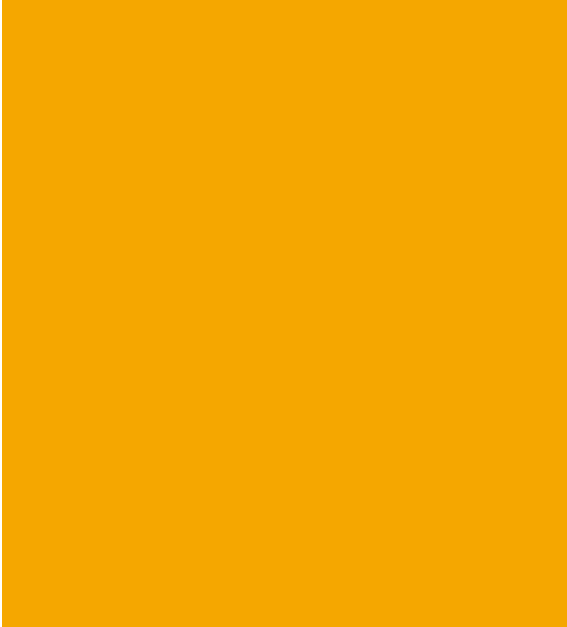
Hierarchical:

- hierarchical clustering views the lower midrange cases as closer to the core bulk





Conclusion





Review

1. Built a reproducible R pipeline to preprocess and cluster five years of World Happiness data
2. Median/MAD scaling combined with a Mahalanobis distance matrix made robust, interpretable clusters
3. k-means and Ward's hierarchical methods produced clear, stable country groupings; DBSCAN was less effective for this task so we did not investigate it more
4. Determined by silhouette scores, correlation, and adjusted Rand indices, This analysis offers a solid, data-driven foundation for targeted policy interventions



Key findings

- Three clear clusters: High, Medium, and Low happiness
- Strong agreement between k-means ($k = 3$) and Ward's hierarchical
- DBSCAN splited data into many small groups. Less useful for broad segmentation.
- 90% of countries stayed in the same cluster from 2015 to 2019
- Robust preprocessing controlled outliers effectively.

Policy hints:

- For Low-happiness cluster: Prioritize economic and social support.
- For High-happiness cluster: Maintain performance via governance and anti-corruption measures.